

Chapitre 9 : L'analyse en composantes principales

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 9 sur *l'analyse en composantes principales* du livre **Langage R et Statistiques : Initiation à l'analyse de données** publié par les éditions ENI (2^{ème} édition).

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.0      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.2      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.1
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```

```
library(FactoMineR)
```

2 Bases mathématiques de l'ACP

2.1 Approximation d'une variable par une autre

fixation de la graine aléatoire

```
set.seed(8)
```

Création des données

```
exemple_relation_lineaire <-
  tibble(
    x = runif(30, min = 0, max = 10),
    y = 3 * x + 5 + runif(30, min = -8, max = 10)
  )
```

réalisation du modèle

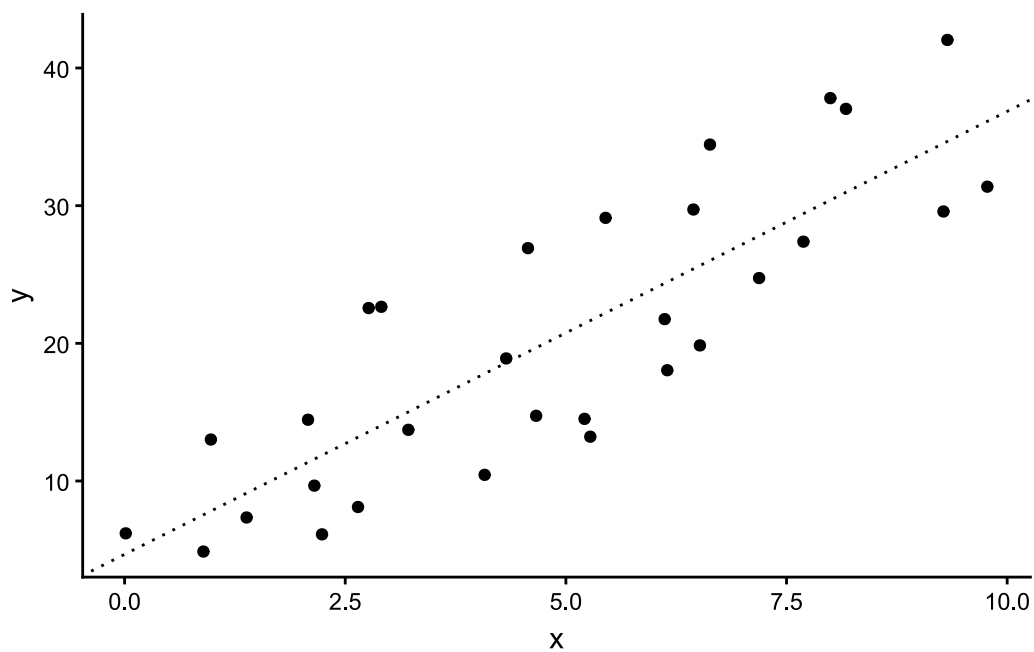
```
modele_lineaire <- lm(y ~ x, exemple_relation_lineaire)
```

représentation graphique

```

exemple_relation_lineaire |>
  ggplot() +
  aes(x = x, y = y) +
  geom_point() +
  theme_classic() +
  geom_abline(
    intercept = modele_lineaire$coefficients[1],
    slope = modele_lineaire$coefficients[2],
    linetype = 3
  )

```



2.2 Ajout de flèches sur la figure précédente pour représenter les dimensions.

Création d'une fonction pour calculer y en fonction de x

```

calcul_y <- function(valeur_x){

  if (!is.numeric(valeur_x)) {stop("la valeur renseignée doit-être un nombre")}

  as.numeric(modele_lineaire$coefficients[1] + modele_lineaire$coefficients[2]
* valeur_x)

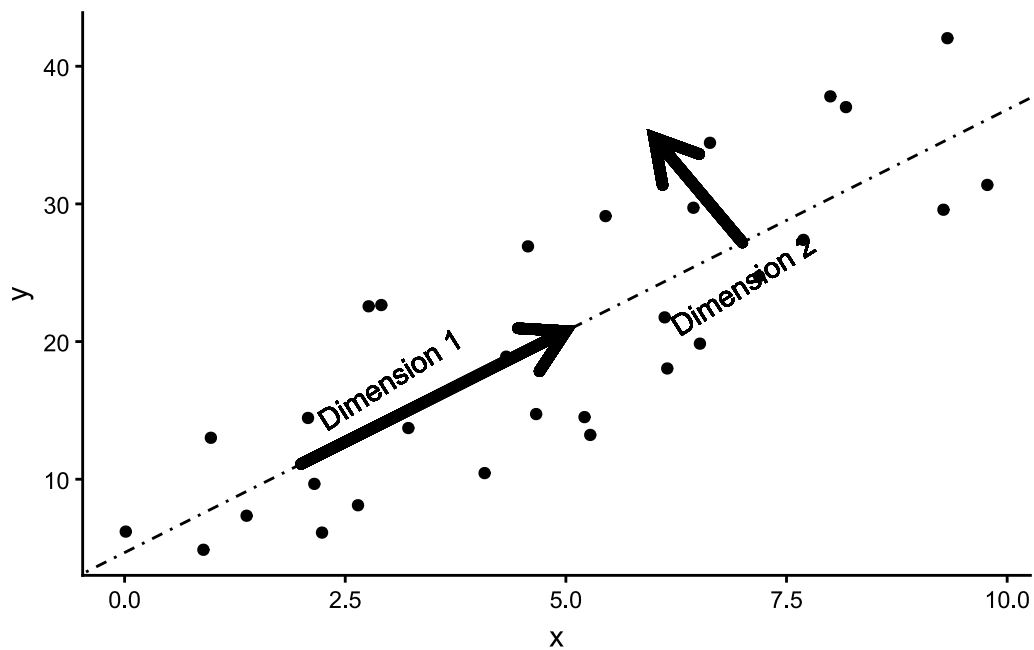
}

```

```

exemple_relation_lineaire |>
  ggplot() +
  aes(x = x, y = y) +
  geom_point() +
  theme_classic() +
  geom_abline(
    intercept = modele_lineaire$coefficients[1],
    slope = modele_lineaire$coefficients[2],
    linetype = 4
  ) +
  geom_segment(
    x = 2,
    y = calcul_y(2),
    xend = 5,
    yend = calcul_y(5),
    arrow = arrow(ends="last"),
    linewidth = 2,
    linejoin = "mitre",
    lineend = "round"
  ) +
  geom_text(
    x = 3,
    y = calcul_y(3) + 3,
    label = "Dimension 1",
    angle = modele_lineaire$coefficients[2] * 10
  ) +
  geom_segment(
    x = 7,
    y = calcul_y(7),
    xend = 6,
    yend = calcul_y(6) * modele_lineaire$coefficients[1]/
modele_lineaire$coefficients[2],
    arrow = arrow(ends="last"),
    linewidth = 2,
    linejoin = "mitre",
    lineend = "round"
  ) +
  geom_text(
    x = 7,
    y = calcul_y(7) - 3,
    label = "Dimension 2",
    angle = modele_lineaire$coefficients[2] * 10
  )

```



2.3 Données centrées-réduites

Le but de cette figure est de montrer que la normalisation des données ne change pas la forme du nuage de points mais la réduction oui.

```
exemple_nuage <- exemple_relation_lineaire |>
mutate_all(
  list(
    centree = ~ scale(.x, center = TRUE, scale = FALSE) |>
      as.numeric(),
    centree_reduite = ~ scale(.x, center = TRUE, scale = TRUE) |>
      as.numeric()
  )
)
```

représentation graphique

```
centre <- exemple_nuage |>
ggplot() +
  geom_point(
    aes(x = x, y = y),
    color = "grey"
  ) +
  geom_point(
    aes(x = x_centree, y = y_centree)
```

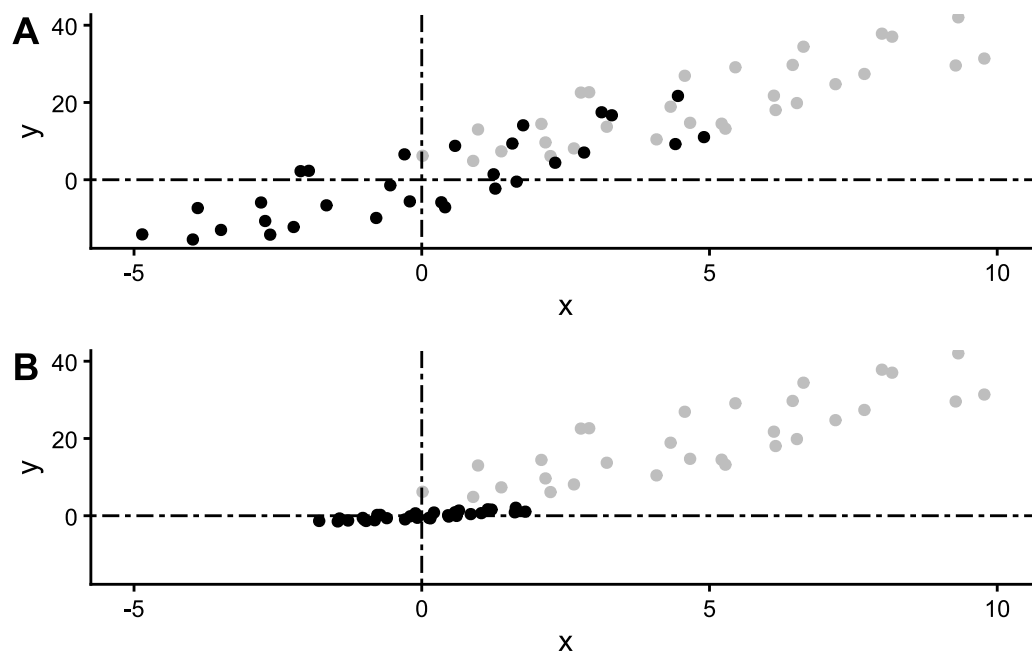
```

    ) +
    geom_hline(aes(yintercept = 0), linetype = "twodash") +
    geom_vline(aes(xintercept = 0), linetype = "twodash") +
    theme_classic() +
    coord_cartesian(ylim = c(-15, 40), xlim = c(-5, 10))

centre_reduit <- exemple_nuage |>
  ggplot() +
  geom_point(
    aes(x = x, y = y),
    color = "grey"
  ) +
  geom_point(
    aes(x = x_centree_reduite, y = y_centree_reduite)
  ) +
  geom_hline(aes(yintercept = 0), linetype = "twodash") +
  geom_vline(aes(xintercept = 0), linetype = "twodash") +
  theme_classic() +
  coord_cartesian(ylim = c(-15, 40), xlim = c(-5, 10))

cowplot::plot_grid(
  centre,
  centre_reduit,
  labels=c("A", "B"),
  ncol = 1,
  nrow = 2)

```



```
ggsave("img/09Rl04N.png", width = 14, height = 14, units = "cm")
```

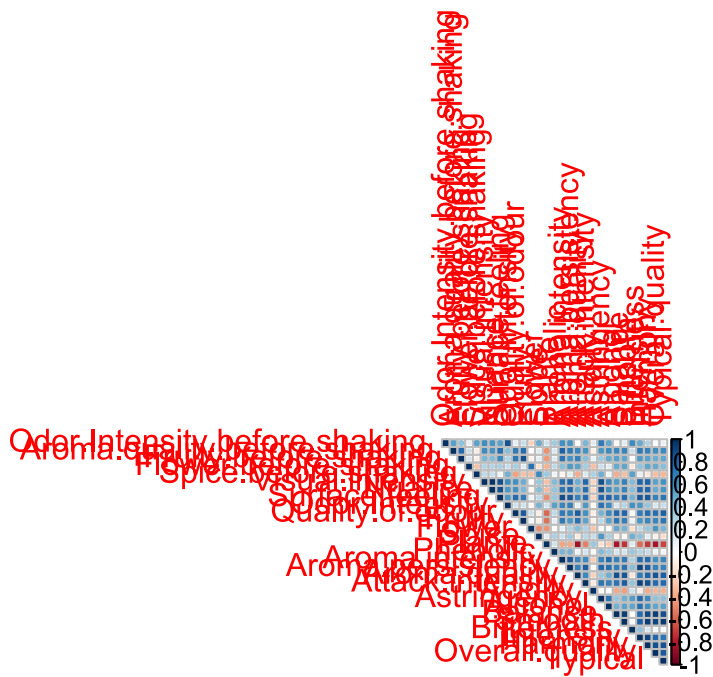
2.4 Matrice de corrélation

chargement des données

```
data("wine")
```

représentation graphique

```
cor(wine |> select(-Label, -Soil)) |> corrplot::corrplot(type = "upper")
```



2.5 Représentation des projections de vecteurs

création des vecteurs

```
vecteurs <- tibble(
  vecteur = c("blue", "red", "gold"),
  A = c(0,0,0),
  B = c(0.4,0.8,-0.1),
  C = c(0.9, -0.3, 0.8),
  type_ligne = c("solid", "dotted", "longdash")
)
```

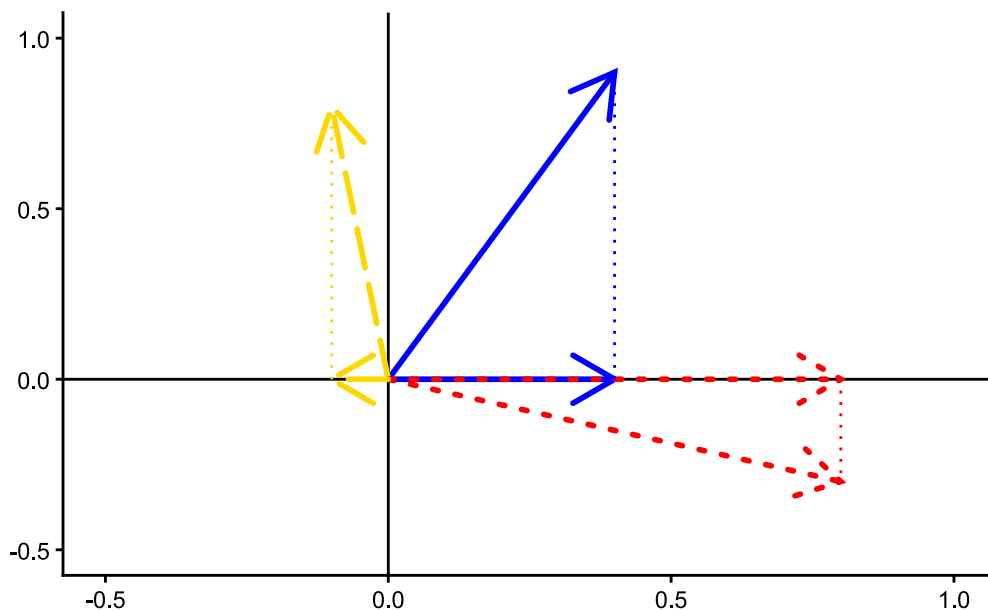
représentation des vecteurs

```
ggplot() +  
  aes(x = -1:1, y = -1:1) +  
  geom_vline(xintercept = 0) +  
  geom_hline(yintercept = 0) +  
  labs(x = "", y = "") +  
  geom_segment(  
    x = vecteurs$A,  
    y = vecteurs$A,  
    xend = vecteurs$B,  
    yend = vecteurs$C,  
    data = vecteurs,
```

```

    color = vecteurs$vecteur,
    arrow = arrow(ends="last"),
    linetype = vecteurs$type_ligne,
    linewidth = 1,
    linejoin = "mitre",
    lineend = "round"
) +
geom_segment(
  x = vecteurs$A,
  y = vecteurs$A,
  xend = vecteurs$B,
  yend = vecteurs$A,
  data = vecteurs,
  color = vecteurs$vecteur,
  arrow = arrow(ends="last"),
  linetype = vecteurs$type_ligne,
  linewidth = 1,
  linejoin = "mitre",
  lineend = "round"
) +
geom_segment(
  x = vecteurs$B,
  y = vecteurs$A,
  xend = vecteurs$B,
  yend = vecteurs$C,
  data = vecteurs,
  color = vecteurs$vecteur,
  linetype = "dotted"
) +
xlim(-0.5,1) +
ylim(-0.5,1) +
theme_classic()

```



```
ggsave("img/08Rl06N.png", width = 14, height = 10, units = "cm")
```

3 Mise en place de l'ACP

3.1 données utilisées

```
data("wine")
glimpse(wine)
```

```
Rows: 21
Columns: 31
$ Label          <fct> Saumur, Saumur, Bourgueuil, Chinon, Saum...
$ Soil           <fct> Env1, Env1, Env1, Env2, Reference, Refer...
$ Odor.Intensity.before.shaking <dbl> 3.074, 2.964, 2.857, 2.808, 3.607, 2.857...
$ Aroma.quality.before.shaking <dbl> 3.000, 2.821, 2.929, 2.593, 3.429, 3.111...
$ Fruity.before.shaking <dbl> 2.714, 2.375, 2.560, 2.417, 3.154, 2.577...
$ Flower.before.shaking <dbl> 2.280, 2.280, 1.960, 1.913, 2.154, 2.040...
$ Spice.before.shaking <dbl> 1.960, 1.680, 2.077, 2.160, 2.040, 2.077...
$ Visual.intensity <dbl> 4.321, 3.222, 3.536, 2.893, 4.393, 4.464...
$ Nuance         <dbl> 4.000, 3.000, 3.393, 2.786, 4.036, 4.259...
$ Surface.feeling <dbl> 3.269, 2.808, 3.000, 2.538, 3.385, 3.407...
$ Odor.Intensity <dbl> 3.407, 3.370, 3.250, 3.160, 3.536, 3.179...
$ Quality.of.odour <dbl> 3.308, 3.000, 2.929, 2.880, 3.360, 3.385...
```

```

$ Fruity          <dbl> 2.885, 2.560, 2.769, 2.391, 3.160, 2.800...
$ Flower          <dbl> 2.320, 2.440, 2.192, 2.083, 2.231, 2.240...
$ Spice           <dbl> 1.840, 1.739, 2.250, 2.167, 2.148, 2.148...
$ Plante          <dbl> 2.000, 2.000, 1.750, 2.304, 1.762, 1.750...
$ Phenolic        <dbl> 1.650, 1.381, 1.250, 1.476, 1.600, 1.476...
$ Aroma.intensity <dbl> 3.259, 2.962, 3.077, 2.542, 3.615, 3.214...
$ Aroma.persistence <dbl> 2.963, 2.808, 2.800, 2.583, 3.296, 3.148...
$ Aroma.quality   <dbl> 3.200, 2.926, 3.077, 2.478, 3.462, 3.321...
$ Attack.intensity <dbl> 2.963, 3.036, 3.222, 2.704, 3.464, 3.286...
$ Acidity         <dbl> 2.107, 2.107, 2.179, 3.179, 2.571, 2.393...
$ Astringency     <dbl> 2.429, 2.179, 2.250, 2.185, 2.536, 2.643...
$ Alcohol         <dbl> 2.500, 2.654, 2.643, 2.500, 2.786, 2.857...
$ Balance         <dbl> 3.250, 2.926, 3.321, 2.333, 3.464, 3.286...
$ Smooth          <dbl> 2.731, 2.500, 2.679, 1.680, 3.036, 2.857...
$ Bitterness      <dbl> 1.926, 1.926, 2.000, 1.963, 2.071, 2.179...
$ Intensity       <dbl> 2.857, 2.893, 3.074, 2.462, 3.643, 3.464...
$ Harmony         <dbl> 3.143, 2.964, 3.143, 2.038, 3.643, 3.500...
$ Overall.quality <dbl> 3.393, 3.214, 3.536, 2.464, 3.741, 3.643...
$ Typical         <dbl> 3.250, 3.036, 3.179, 2.250, 3.444, 3.393...

```

```
summary(wine)
```

Label	Soil	Odor.Intensity.before.shaking
Saumur :11	Reference:7	Min. :2.643
Bourgueuil: 6	Env1 :7	1st Qu.:2.893
Chinon : 4	Env2 :5	Median :3.071
	Env4 :2	Mean :3.111
		3rd Qu.:3.214
		Max. :3.708
Aroma.quality.before.shaking Fruity.before.shaking Flower.before.shaking		
Min. :2.593		Min. :2.375
1st Qu.:2.926		1st Qu.:2.560
Median :3.107		Median :2.731
Mean :3.046		Mean :2.714
3rd Qu.:3.179		3rd Qu.:2.833
Max. :3.429		Max. :3.154
Spice.before.shaking Visual.intensity Nuance Surface.feeling		
Min. :1.667	Min. :2.607	Min. :2.536
1st Qu.:1.808	1st Qu.:3.704	1st Qu.:3.407
Median :1.962	Median :4.071	Median :3.893
Mean :1.991	Mean :3.958	Mean :3.727
3rd Qu.:2.077	3rd Qu.:4.321	3rd Qu.:4.036

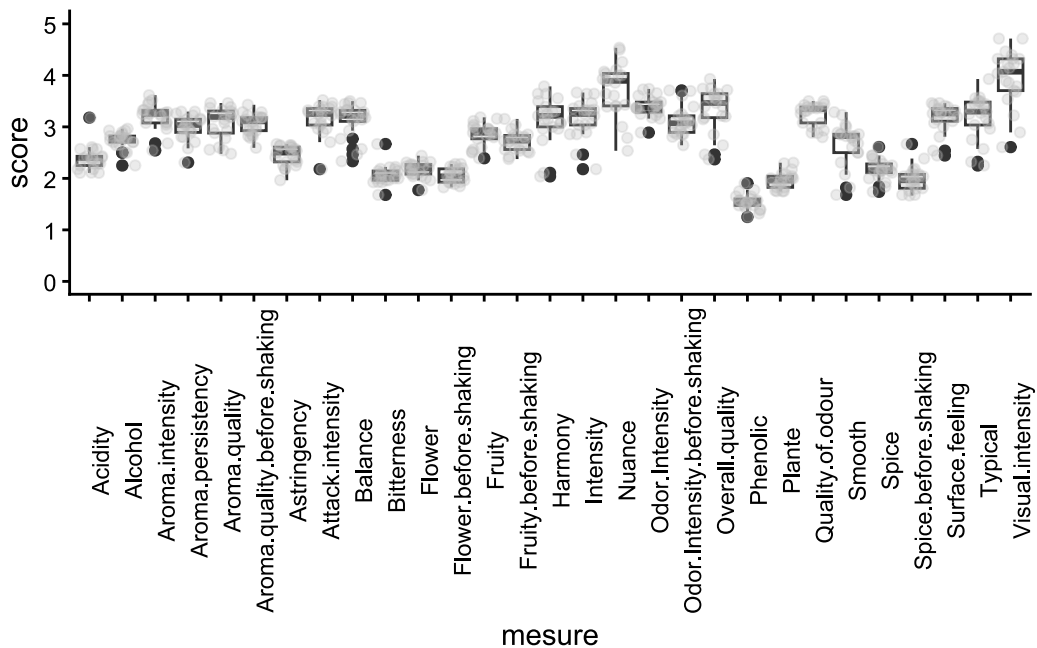
Max. :2.667	Max. :4.714	Max. :4.536	Max. :3.462
Odor.Intensity	Quality.of.odour	Fruity	Flower
Min. :2.889	Min. :2.800	Min. :2.391	Min. :1.773
1st Qu.:3.286	1st Qu.:3.077	1st Qu.:2.769	1st Qu.:2.083
Median :3.370	Median :3.346	Median :2.885	Median :2.192
Mean :3.395	Mean :3.237	Mean :2.847	Mean :2.162
3rd Qu.:3.481	3rd Qu.:3.385	3rd Qu.:2.963	3rd Qu.:2.269
Max. :3.737	Max. :3.500	Max. :3.185	Max. :2.440
Spice	Plante	Phenolic	Aroma.intensity
Min. :1.739	Min. :1.750	Min. :1.250	Min. :2.542
1st Qu.:2.111	1st Qu.:1.826	1st Qu.:1.476	1st Qu.:3.077
Median :2.160	Median :1.957	Median :1.545	Median :3.250
Mean :2.178	Mean :1.964	Mean :1.542	Mean :3.197
3rd Qu.:2.280	3rd Qu.:2.042	3rd Qu.:1.600	3rd Qu.:3.321
Max. :2.609	Max. :2.304	Max. :1.905	Max. :3.615
Aroma.persistency	Aroma.quality	Attack.intensity	Acidity
Min. :2.308	Min. :2.478	Min. :2.179	Min. :2.107
1st Qu.:2.893	1st Qu.:2.880	1st Qu.:3.036	1st Qu.:2.250
Median :3.036	Median :3.192	Median :3.250	Median :2.393
Mean :2.983	Mean :3.064	Mean :3.156	Mean :2.385
3rd Qu.:3.148	3rd Qu.:3.308	3rd Qu.:3.357	3rd Qu.:2.429
Max. :3.296	Max. :3.462	Max. :3.519	Max. :3.179
Astringency	Alcohol	Balance	Smooth
Min. :1.964	Min. :2.250	Min. :2.333	Min. :1.680
1st Qu.:2.321	1st Qu.:2.704	1st Qu.:3.107	1st Qu.:2.500
Median :2.500	Median :2.778	Median :3.214	Median :2.821
Mean :2.440	Mean :2.741	Mean :3.129	Mean :2.674
3rd Qu.:2.607	3rd Qu.:2.821	3rd Qu.:3.321	3rd Qu.:2.857
Max. :2.667	Max. :2.963	Max. :3.500	Max. :3.286
Bitterness	Intensity	Harmony	Overall.quality
Min. :1.679	Min. :2.179	Min. :2.038	Min. :2.370
1st Qu.:1.964	1st Qu.:3.037	1st Qu.:3.000	1st Qu.:3.179
Median :2.037	Median :3.250	Median :3.214	Median :3.464
Mean :2.068	Mean :3.166	Mean :3.148	Mean :3.331
3rd Qu.:2.143	3rd Qu.:3.357	3rd Qu.:3.393	3rd Qu.:3.643
Max. :2.667	Max. :3.667	Max. :3.786	Max. :3.929
Typical			
Min. :2.250			
1st Qu.:3.036			
Median :3.296			
Mean :3.197			
3rd Qu.:3.481			
Max. :3.929			

3.2 Sélection des variables actives

```
vins <- wine |>
  select(-Label, -Soil)
```

3.3 Réduction des variables

```
vins |>
  rownames_to_column("Identification_vin") |>
  pivot_longer(
    !Identification_vin,
    names_to = "measure",
    values_to = "score"
  ) |>
  ggplot() +
    aes(x = measure, y = score) +
    geom_boxplot() +
    geom_jitter(color = "grey", alpha = 0.3) +
    theme_classic() +
    theme(axis.text.x = element_text(angle=90)) +
    coord_cartesian(ylim = c(0,5))
```



```
ggsave(
  "img/09Rl07N.png", width = 14, units = "cm"
)
```

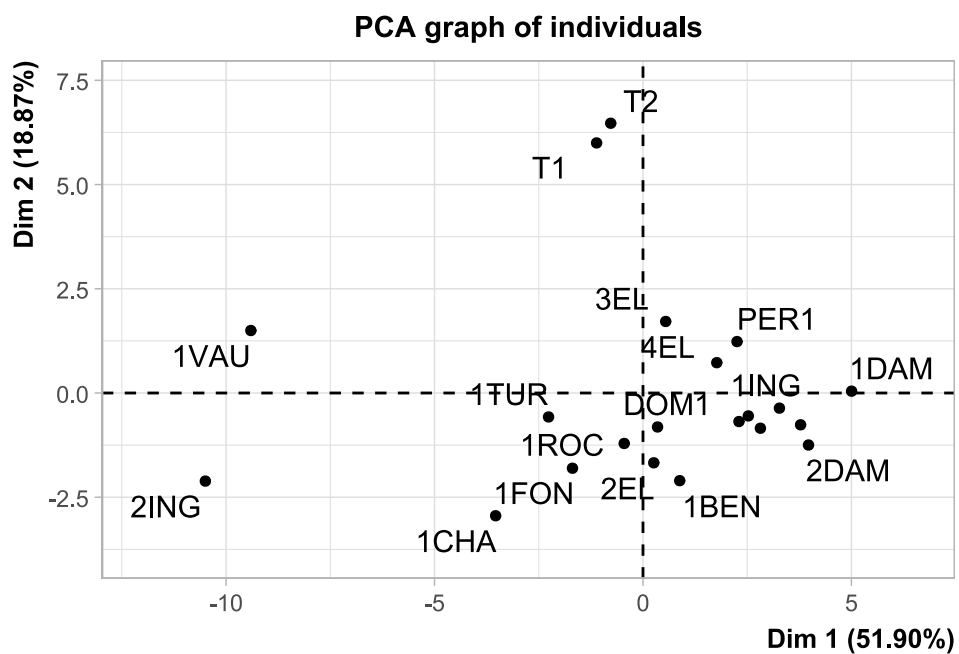
Saving 14 x 8.89 cm image

3.4 Réalisation de l'ACP

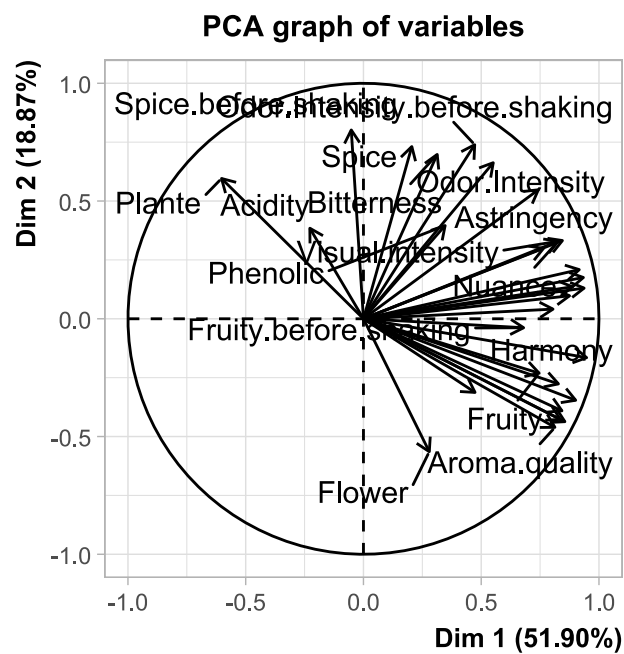
création de l'acp sur le jeu de données vins

```
acp <- PCA(X = vins)
```

Warning: ggrepel: 4 unlabeled data points (too many overlaps). Consider increasing max.overlaps



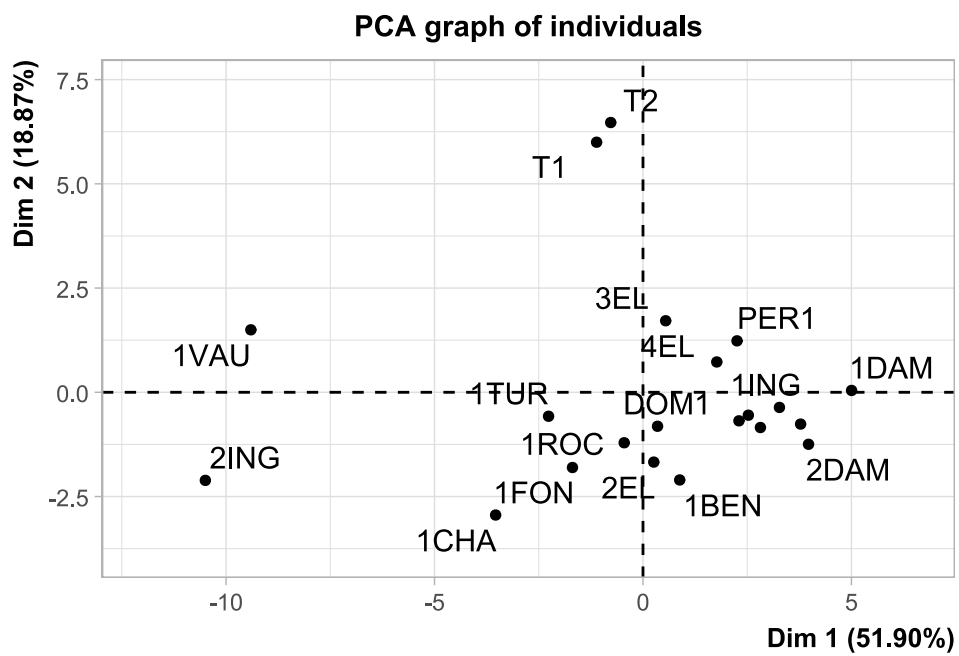
Warning: ggrepel: 13 unlabeled data points (too many overlaps). Consider increasing max.overlaps



afficher à nouveau le graphique des individus suite à la réalisation de l'ACP

```
plot.PCA(acp, choix = "ind")
```

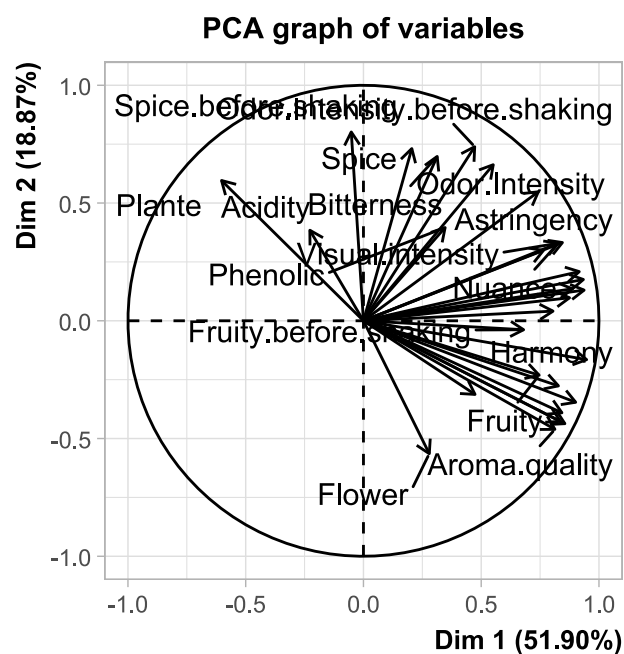
Warning: ggrepel: 4 unlabeled data points (too many overlaps). Consider increasing max.overlaps



afficher à nouveau le graphique des variables suite à la réalisation de l'ACP

```
plot.PCA(acp, choix = "var")
```

Warning: ggrepel: 13 unlabeled data points (too many overlaps). Consider increasing max.overlaps



3.5 Pourcentage d'inertie de toutes les dimensions

```
acp$eig
```

	eigenvalue	percentage of variance	cumulative percentage of variance
comp 1	15.04989130	51.89617688	51.89618
comp 2	5.47180980	18.86830964	70.76449
comp 3	2.01715336	6.95570124	77.72019
comp 4	1.35956240	4.68814620	82.40833
comp 5	1.10129223	3.79755940	86.20589
comp 6	0.81561992	2.81248248	89.01838
comp 7	0.78431637	2.70453921	91.72292
comp 8	0.55810712	1.92450731	93.64742
comp 9	0.40614877	1.40051301	95.04794
comp 10	0.37205590	1.28295138	96.33089
comp 11	0.30777124	1.06128014	97.39217
comp 12	0.22130583	0.76312357	98.15529
comp 13	0.12504345	0.43118433	98.58647
comp 14	0.12165224	0.41949048	99.00597
comp 15	0.09202732	0.31733559	99.32330
comp 16	0.06510431	0.22449761	99.54780
comp 17	0.04371431	0.15073901	99.69854
comp 18	0.04144372	0.14290939	99.84145

comp 19	0.02987389	0.10301340	99.94446
comp 20	0.01610652	0.05553974	100.00000

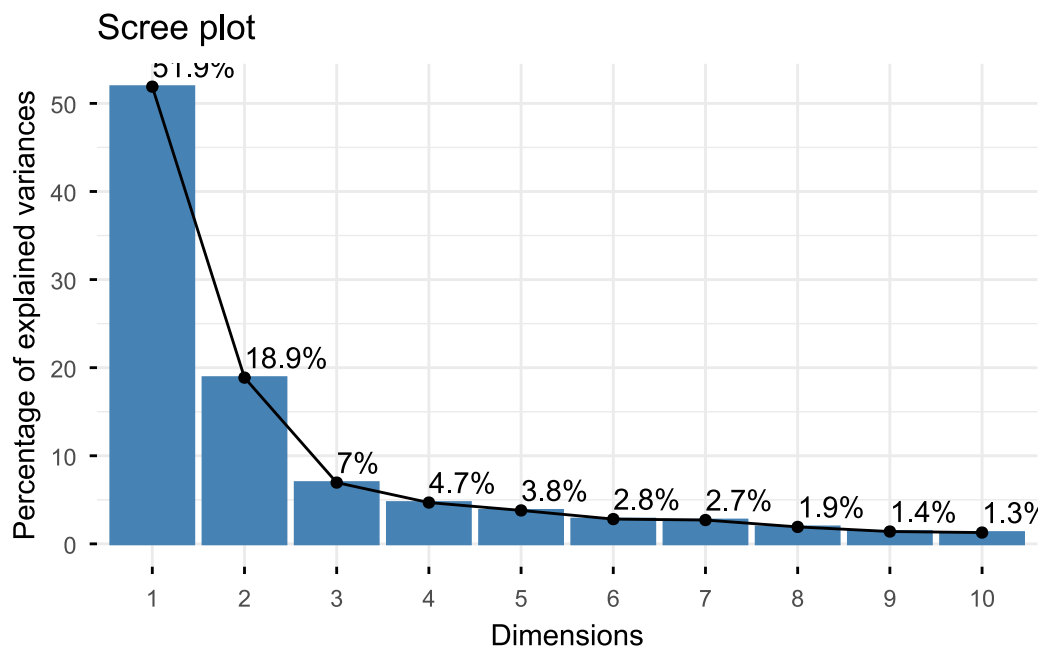
représentation du pourcentage d'inertie

```
library(factoextra)
```

Welcome! Want to learn more? See two factoextra-related books at <https://goo.gl/ve3WBa>

```
fviz_screplot(acp, addlabels = TRUE)
```

Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
Ignoring empty aesthetic: `width`.



3.6 Fonction `dimdesc()`

```
dimdesc(acp)$Dim.1
```

Link between the variable and the continuous variables (R-square)

	correlation	p.value
Harmony	0.9477389	7.097145e-11
Intensity	0.9370041	4.013113e-10
Surface.feeling	0.9339786	6.191964e-10
Aroma.persistency	0.9163801	5.450361e-09
Smooth	0.9012129	2.499196e-08
Aroma.intensity	0.8874982	8.132041e-08
Attack.intensity	0.8751516	2.080292e-07
Balance	0.8554023	7.745878e-07
Visual.intensity	0.8448838	1.445582e-06
Typical	0.8428752	1.620078e-06
Overall.quality	0.8360574	2.358197e-06
Quality.of.odour	0.8280174	3.594771e-06
Nuance	0.8268412	3.816584e-06
Aroma.quality	0.8130238	7.472662e-06
Aroma.quality.before.shaking	0.8049316	1.080432e-05
Alcohol	0.7644194	5.468104e-05
Fruity	0.7460760	1.030624e-04
Astringency	0.7425775	1.156169e-04
Fruity.before.shaking	0.6801860	6.919600e-04
Odor.Intensity	0.5510875	9.617824e-03
Flower.before.shaking	0.4738594	3.000516e-02
Odor.Intensity.before.shaking	0.4723474	3.060745e-02
Plante	-0.6027254	3.829791e-03

```
dimdesc(acp)$Dim.2
```

Link between the variable and the continuous variables (R-square)

	correlation	p.value
Spice.before.shaking	0.8017349	1.244078e-05
Odor.Intensity.before.shaking	0.7391855	1.290281e-04
Spice	0.7312254	1.658783e-04
Bitterness	0.6978365	4.361867e-04
Odor.Intensity	0.6626751	1.062158e-03
Plante	0.5956395	4.384868e-03
Astringency	0.5477188	1.016319e-02
Balance	-0.4369098	4.765885e-02
Aroma.quality	-0.4610135	3.543125e-02
Flower	-0.5644466	7.685201e-03

3.7 Choix du nombre de dimensions

```
FactoInvestigate::eigenRef(acp)
```

```
$datasets
[1] 10000

$quantile
[1] 0.95

$inertia
      dim.1      dim.2      dim.3      dim.4      dim.5      dim.6      dim.7      dim.8
0.1611370 0.2886861 0.3975124 0.4924203 0.5751173 0.6475322 0.7099723 0.7644784
      dim.9      dim.10      dim.11      dim.12      dim.13      dim.14      dim.15      dim.16
0.8110514 0.8512241 0.8854971 0.9138062 0.9370109 0.9558067 0.9705800 0.9818416
      dim.17      dim.18      dim.19      dim.20
0.9900770 0.9956030 0.9988644 1.0000000
```

```
FactoInvestigate::inertiaDistrib(acp)
```

L'inertie des axes factoriels indique d'une part si les variables sont structurées et suggère d'autre part le nombre judicieux de composantes principales à étudier.

Les 2 premiers axes de l'analyse expriment **70.76%** de l'inertie totale du jeu de données ; cela signifie que 70.76% de la variabilité totale du nuage des individus (ou des variables) est représentée dans ce plan.

C'est un pourcentage élevé, et le premier plan représente donc bien la variabilité contenue dans une très large part du jeu de données actif.

Cette valeur est nettement supérieure à la valeur référence de **28.85%**, la variabilité expliquée par ce plan est donc hautement significative

(cette inertie de référence est le quantile 0.95-quantile de la distribution des pourcentages d'inertie obtenue en simulant 10000 jeux de données aléatoires de dimensions comparables sur la base d'une distribution normale).

Du fait de ces observations, il n'est probablement pas nécessaire pour l'analyse d'interpréter les dimensions suivantes.

```
...

par(mar = c(2.6, 4.1, 1.1, 2.1))
ggplot2::ggplot(cbind.data.frame(x=1:nrow(res$eig),y=res$eig[,2]))
+   ggplot2::aes(x=x,      y=y)+   ggplot2::geom_col(fill="blue")      +
ggplot2::xlab("Dimension")      +   ggplot2::ylab("Pourcentage d'inertie")      +
```

```
ggplot2::ggtitle("Décomposition de l'inertie totale") + ggplot2::theme_light()
+ ggplot2::theme(plot.title = ggplot2::element_text(hjust =0.5)) +
ggplot2::scale_x_continuous(breaks=1:nrow(res$eig))
```

```

**\*\*Figure - Décomposition de l'inertie totale\*\***

\*Le premier facteur est prépondérant : il explique à lui seul 51.9% de la variabilité totale des données.\*

\*Il convient de noter que dans un tel cas, la variabilité liée aux autres composantes peut être dénuée de sens, en dépit d'un pourcentage élevé.\*

Une estimation du nombre pertinent d'axes à interpréter suggère de restreindre l'analyse à la description des 2 premiers axes.

Ces composantes révèlent un taux d'inertie supérieur à celle du quantile 0.95-quantile de distributions aléatoires (70.76% contre 28.85%).

Cette observation suggère que seuls ces axes sont porteurs d'une véritable information.

En conséquence, la description de l'analyse sera restreinte à ces seuls axes.

```
[1] 2
```

```
acp$eig
```

|         | eigenvalue  | percentage of variance | cumulative percentage of variance |
|---------|-------------|------------------------|-----------------------------------|
| comp 1  | 15.04989130 | 51.89617688            | 51.89618                          |
| comp 2  | 5.47180980  | 18.86830964            | 70.76449                          |
| comp 3  | 2.01715336  | 6.95570124             | 77.72019                          |
| comp 4  | 1.35956240  | 4.68814620             | 82.40833                          |
| comp 5  | 1.10129223  | 3.79755940             | 86.20589                          |
| comp 6  | 0.81561992  | 2.81248248             | 89.01838                          |
| comp 7  | 0.78431637  | 2.70453921             | 91.72292                          |
| comp 8  | 0.55810712  | 1.92450731             | 93.64742                          |
| comp 9  | 0.40614877  | 1.40051301             | 95.04794                          |
| comp 10 | 0.37205590  | 1.28295138             | 96.33089                          |
| comp 11 | 0.30777124  | 1.06128014             | 97.39217                          |
| comp 12 | 0.22130583  | 0.76312357             | 98.15529                          |
| comp 13 | 0.12504345  | 0.43118433             | 98.58647                          |
| comp 14 | 0.12165224  | 0.41949048             | 99.00597                          |
| comp 15 | 0.09202732  | 0.31733559             | 99.32330                          |
| comp 16 | 0.06510431  | 0.22449761             | 99.54780                          |
| comp 17 | 0.04371431  | 0.15073901             | 99.69854                          |

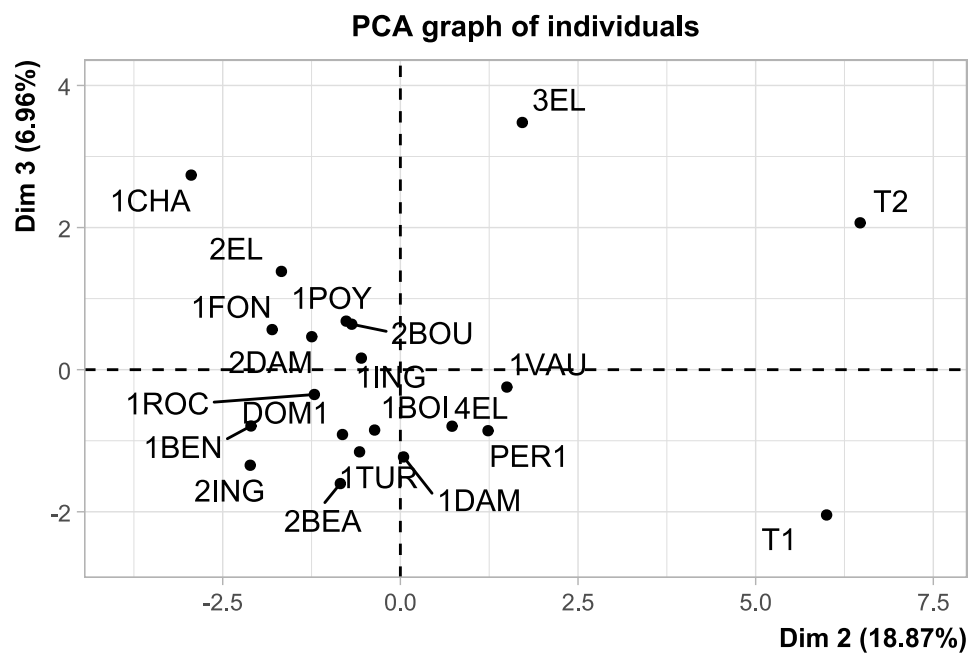
|         |            |            |           |
|---------|------------|------------|-----------|
| comp 18 | 0.04144372 | 0.14290939 | 99.84145  |
| comp 19 | 0.02987389 | 0.10301340 | 99.94446  |
| comp 20 | 0.01610652 | 0.05553974 | 100.00000 |

### 3.8 ACP avec le nombre de dimensions à interpréter fixé

```
acp <- PCA(
 X = vins,
 ncp = 3,
 graph = FALSE
)
```

### 3.9 Vérification troisième dimension

```
plot.PCA(acp, axes = c(2, 3), choix = "ind")
```



### 3.10 Vérification des individus extremes

```
acp_sans_extreme <- PCA(
 X = setdiff(vins, vins[c("1VAU", "1ING"),]),
 ncp = 3,
 graph = FALSE
)
```

```
acp_sans_extreme$eig
```

|         | eigenvalue  | percentage of variance | cumulative percentage of variance |
|---------|-------------|------------------------|-----------------------------------|
| comp 1  | 13.67794907 | 47.16534161            | 47.16534                          |
| comp 2  | 6.16613308  | 21.26252786            | 68.42787                          |
| comp 3  | 2.50112143  | 8.62455665             | 77.05243                          |
| comp 4  | 1.54945028  | 5.34293200             | 82.39536                          |
| comp 5  | 1.13711730  | 3.92109415             | 86.31645                          |
| comp 6  | 0.87638632  | 3.02202180             | 89.33847                          |
| comp 7  | 0.78041122  | 2.69107317             | 92.02955                          |
| comp 8  | 0.49421605  | 1.70419328             | 93.73374                          |
| comp 9  | 0.37384950  | 1.28913622             | 95.02288                          |
| comp 10 | 0.34257250  | 1.18128448             | 96.20416                          |
| comp 11 | 0.30054137  | 1.03634954             | 97.24051                          |
| comp 12 | 0.25376934  | 0.87506669             | 98.11558                          |
| comp 13 | 0.15466737  | 0.53333577             | 98.64891                          |
| comp 14 | 0.14263313  | 0.49183837             | 99.14075                          |
| comp 15 | 0.10878824  | 0.37513187             | 99.51588                          |
| comp 16 | 0.06896125  | 0.23779742             | 99.75368                          |
| comp 17 | 0.04611241  | 0.15900833             | 99.91269                          |
| comp 18 | 0.02532014  | 0.08731081             | 100.00000                         |

```
dimdesc(acp_sans_extreme)$Dim.1
```

Link between the variable and the continuous variables (R-square)

|                   | correlation | p.value      |
|-------------------|-------------|--------------|
| Intensity         | 0.9322051   | 6.368818e-09 |
| Aroma.persistency | 0.9267217   | 1.210450e-08 |
| Harmony           | 0.9160683   | 3.698134e-08 |
| Surface.feeling   | 0.9157716   | 3.806822e-08 |
| Attack.intensity  | 0.8810943   | 6.318947e-07 |
| Smooth            | 0.8414309   | 6.338451e-06 |
| Aroma.intensity   | 0.8396105   | 6.938769e-06 |
| Visual.intensity  | 0.8113279   | 2.491208e-05 |
| Quality.of.odour  | 0.7964456   | 4.498794e-05 |
| Nuance            | 0.7928710   | 5.148395e-05 |
| Balance           | 0.7742663   | 9.987302e-05 |
| Typical           | 0.7711851   | 1.108014e-04 |

|                              |           |              |
|------------------------------|-----------|--------------|
| Overall.quality              | 0.7623353 | 1.480382e-04 |
| Alcohol                      | 0.7470398 | 2.376005e-04 |
| Aroma.quality                | 0.7468889 | 2.386729e-04 |
| Aroma.quality.before.shaking | 0.7410576 | 2.833977e-04 |
| Astringency                  | 0.7338702 | 3.481253e-04 |
| Fruity                       | 0.6785455 | 1.404064e-03 |
| Fruity.before.shaking        | 0.6337528 | 3.573656e-03 |
| Odor.Intensity               | 0.5297216 | 1.966737e-02 |

```
dimdesc(acp_sans_extreme)$Dim.2
```

Link between the variable and the continuous variables (R-square)

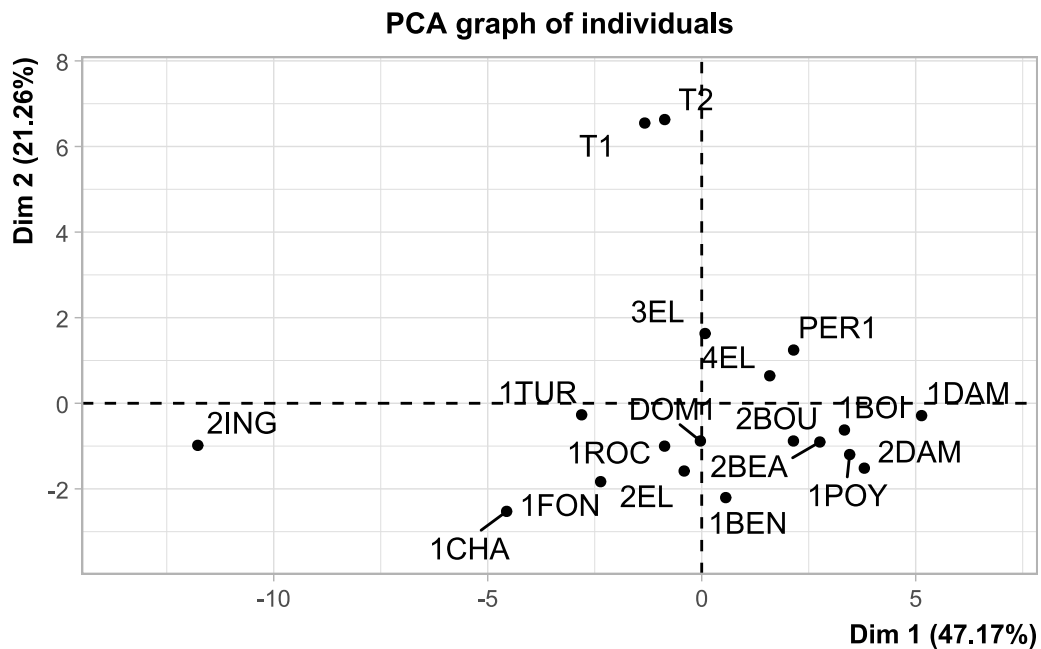
|                               | correlation | p.value      |
|-------------------------------|-------------|--------------|
| Spice.before.shaking          | 0.7774716   | 8.949478e-05 |
| Odor.Intensity.before.shaking | 0.7610924   | 1.540362e-04 |
| Spice                         | 0.7075621   | 7.022128e-04 |
| Plante                        | 0.7026080   | 7.948359e-04 |
| Bitterness                    | 0.6900287   | 1.077205e-03 |
| Odor.Intensity                | 0.6705994   | 1.675668e-03 |
| Astringency                   | 0.5539964   | 1.384955e-02 |
| Typical                       | -0.4821213  | 3.658099e-02 |
| Overall.quality               | -0.5283993  | 2.003259e-02 |
| Balance                       | -0.5493818  | 1.483318e-02 |
| Aroma.quality                 | -0.5521675  | 1.423293e-02 |
| Flower                        | -0.6001848  | 6.591270e-03 |

```
dimdesc(acp_sans_extreme)$Dim.3
```

Link between the variable and the continuous variables (R-square)

|                       | correlation | p.value     |
|-----------------------|-------------|-------------|
| Flower                | 0.5571567   | 0.013206571 |
| Flower.before.shaking | 0.5147048   | 0.024139419 |
| Acidity               | -0.5765771  | 0.009761341 |
| Fruity                | -0.6138165  | 0.005182249 |
| Fruity.before.shaking | -0.6512460  | 0.002524919 |

```
plot.PCA(acp_sans_extreme, choix = "ind")
```



### 3.11 Ajout de variables supplémentaires

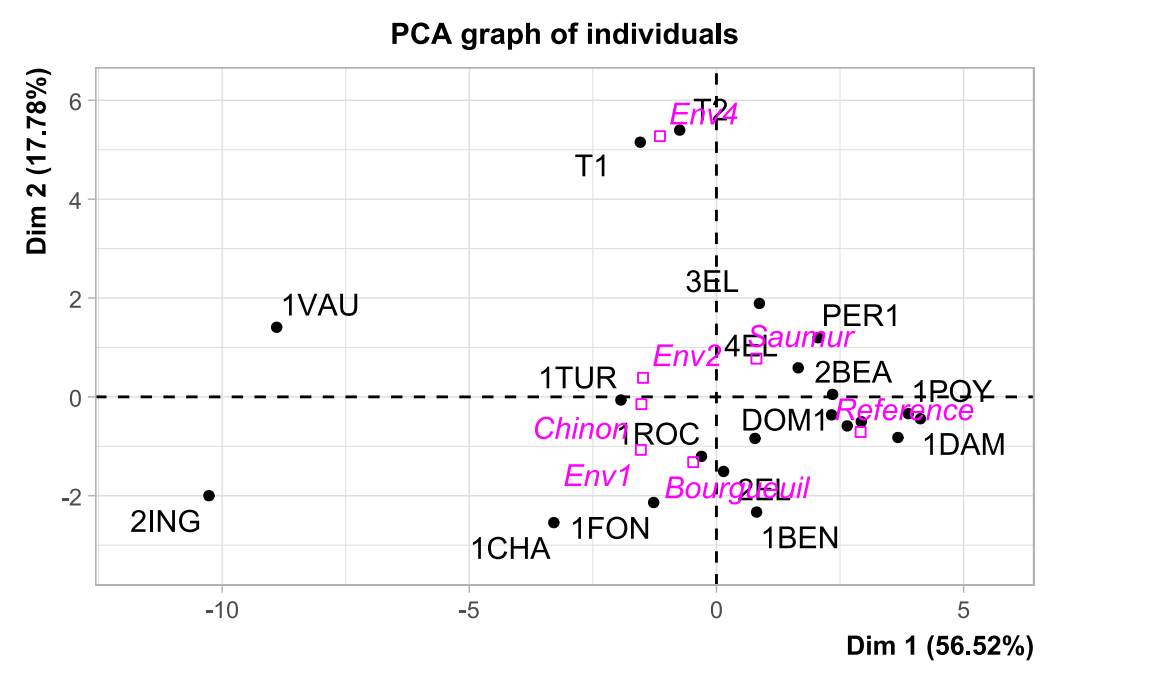
a réaliser sur le jeu de données d'origine `wine`

```
acp_complete <- PCA(
 X = wine,
 ncp = 3,
 quali.sup = c(1,2),
 quanti.sup =
 (
 names(wine) |>
 stringr::str_locate("before") |>
 tibble::as_tibble() |>
 rowid_to_column() |>
 filter(!is.na(start))
)$rowid,
 graph = FALSE
)
```

nuage des individus

```
plot.PCA(acp_complete, choix = "ind")
```

```
Warning: ggrepel: 4 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



```
ggsave(
 "img/09Rl18N.png", width = 14, units = "cm"
)
```

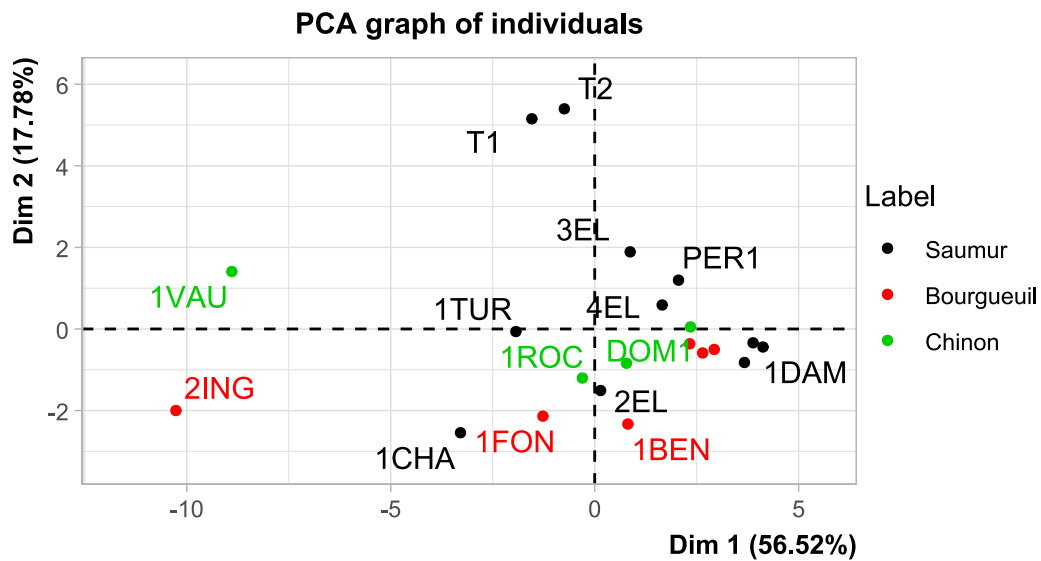
Saving 14 x 8.89 cm image

```
Warning: ggrepel: 2 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

choix de l'habillage `Label`

```
plot.PCA(acp_complete, choix = "ind", habillage = 1, invisible = "quali")
```

```
Warning: ggrepel: 6 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



```
ggsave(
 "img/09Rl19N.png", width = 14, units = "cm"
)
```

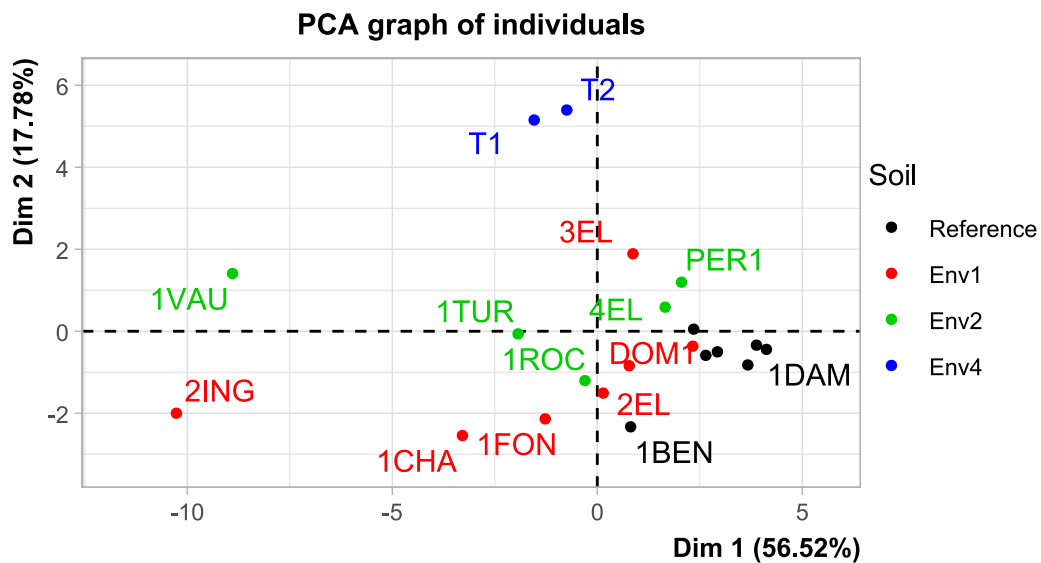
Saving 14 x 8.89 cm image

Warning: ggrepel: 6 unlabeled data points (too many overlaps). Consider increasing max.overlaps

choix de l'habillage Soil

```
plot.PCA(acp_complete, choix = "ind", habillage = 2, invisible = "quali")
```

Warning: ggrepel: 6 unlabeled data points (too many overlaps). Consider increasing max.overlaps



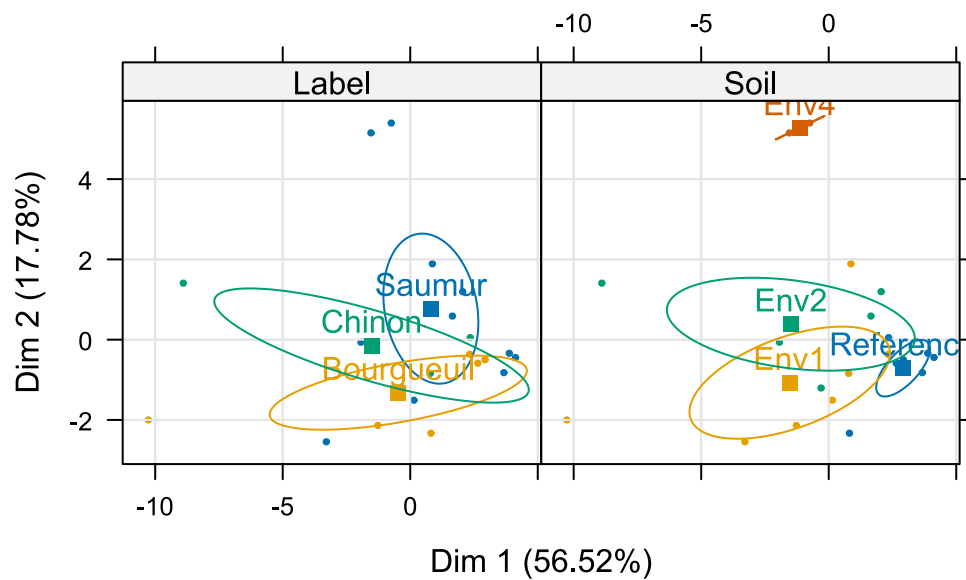
```
ggsave(
 "img/09Rl20N.png", width = 14, units = "cm"
)
```

Saving 14 x 8.89 cm image

Warning: ggrepel: 6 unlabeled data points (too many overlaps). Consider increasing max.overlaps

traçage des ellipses de confiance autours des modalités

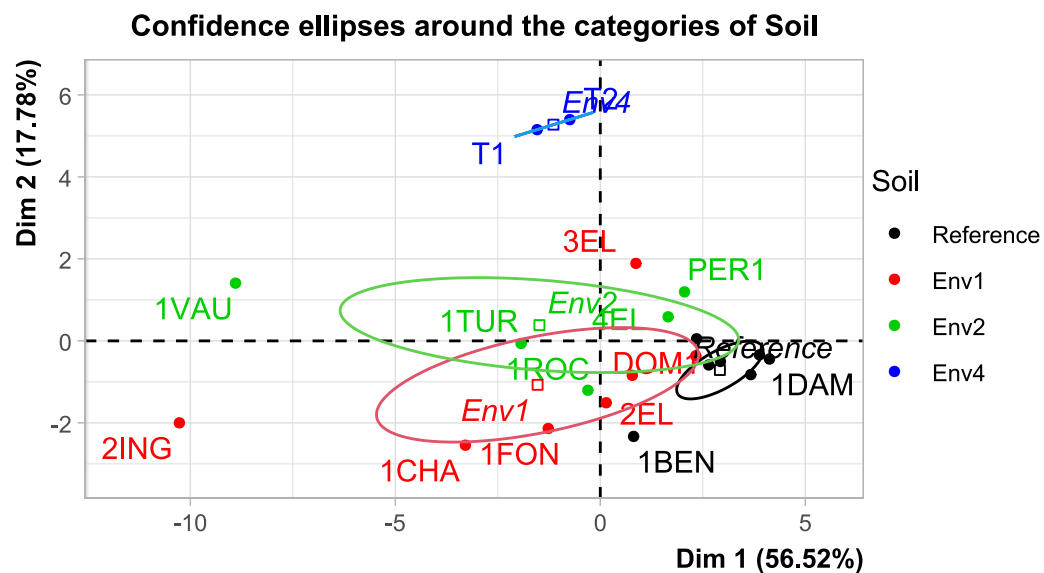
```
plotellipses(acp_complete)
```



seulement la variable Soil

```
plotellipses(acp_complete, keepvar = "Soil")
```

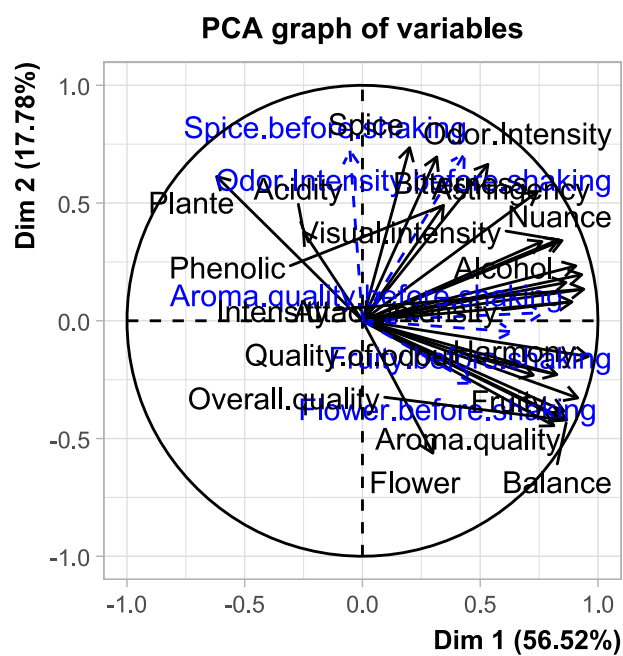
Warning: ggrepel: 6 unlabeled data points (too many overlaps). Consider increasing max.overlaps



nuage des variables

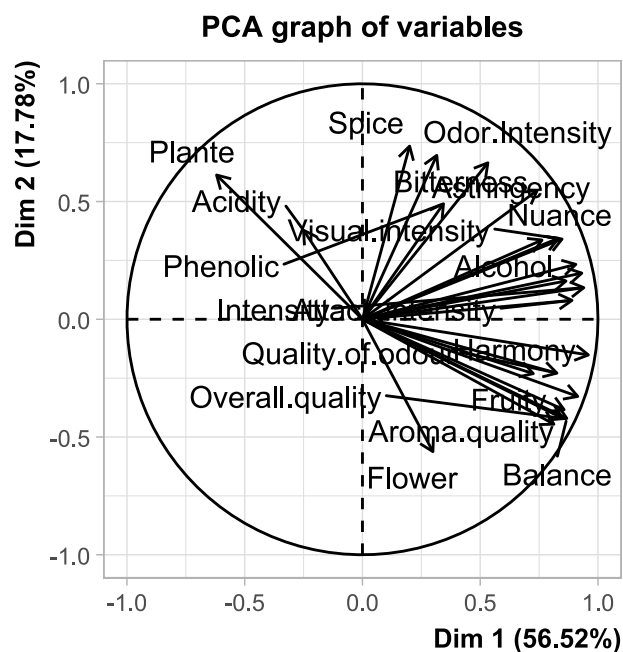
```
plot.PCA(acp_complete, choix = "var")
```

Warning: ggrepel: 5 unlabeled data points (too many overlaps). Consider increasing max.overlaps

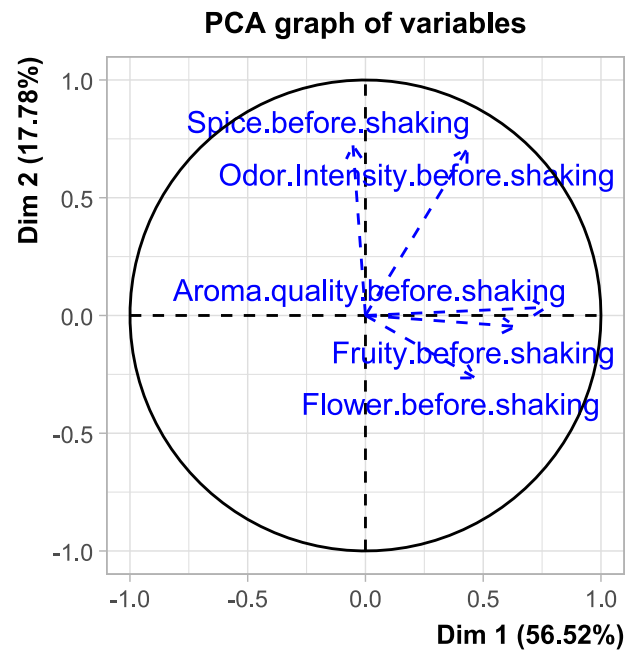


```
plot.PCA(acp_complete, choix = "var", invisible = "quanti.sup")
```

Warning: ggrepel: 5 unlabeled data points (too many overlaps). Consider increasing max.overlaps



```
plot.PCA(acp_complete, choix = "var", invisible = "var")
```



## 4 Cas des données manquantes

```
library(missMDA)
```

import et visualisation des données

```
data("snorena")
summary(snorena)
```

| age     |        | weight  |        | size    |        | alcohol |         | sex  |     |
|---------|--------|---------|--------|---------|--------|---------|---------|------|-----|
| Min.    | :23.00 | Min.    | : 42.0 | Min.    | :158.0 | Min.    | : 0.000 | M    | :75 |
| 1st Qu. | :43.00 | 1st Qu. | : 77.0 | 1st Qu. | :166.0 | 1st Qu. | : 0.000 | W    | :22 |
| Median  | :51.00 | Median  | : 94.0 | Median  | :186.0 | Median  | : 2.000 | NA's | : 3 |
| Mean    | :52.16 | Mean    | : 90.4 | Mean    | :181.1 | Mean    | : 2.905 |      |     |
| 3rd Qu. | :63.00 | 3rd Qu. | :104.5 | 3rd Qu. | :194.0 | 3rd Qu. | : 4.000 |      |     |
| Max.    | :74.00 | Max.    | :120.0 | Max.    | :208.0 | Max.    | :15.000 |      |     |
| NA's    | :3     | NA's    | :9     | NA's    | :5     | NA's    | :5      |      |     |
| snore   |        | tobacco |        |         |        |         |         |      |     |
| N       | :62    | N       | :32    |         |        |         |         |      |     |
| Y       | :32    | Y       | :61    |         |        |         |         |      |     |
| NA's    | : 6    | NA's    | : 7    |         |        |         |         |      |     |

visualisation des données pour lesquels il n'y a pas de données dans la colonne weight

```
snorena |>
 filter(is.na(weight))
```

|   | age | weight | size | alcohol | sex | snore | tobacco |
|---|-----|--------|------|---------|-----|-------|---------|
| 1 | 56  | NA     | 164  | 7       | M   | Y     | <NA>    |
| 2 | 46  | NA     | 208  | 3       | M   | N     | Y       |
| 3 | 46  | NA     | 188  | 0       | W   | N     | N       |
| 4 | 40  | NA     | 193  | 5       | M   | N     | Y       |
| 5 | 55  | NA     | 194  | 1       | M   | N     | <NA>    |
| 6 | 46  | NA     | 196  | 0       | M   | N     | Y       |
| 7 | 64  | NA     | 164  | 4       | M   | Y     | Y       |
| 8 | 62  | NA     | 165  | 0       | M   | N     | Y       |
| 9 | 57  | NA     | 180  | 5       | M   | N     | Y       |

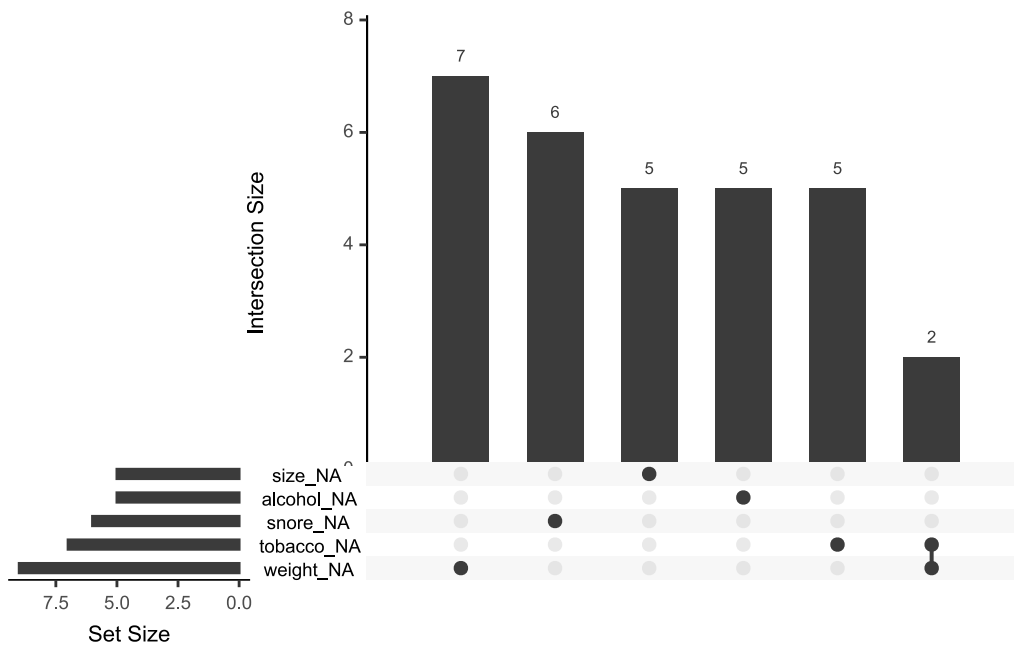
représentation graphique des valeurs manquantes

```
naniar::gg_miss_upset(snorena)
```

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the UpSetR package.
Please report the issue to the authors.
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the UpSetR package.
Please report the issue to the authors.
```

```
Warning: The `size` argument of `element_line()` is deprecated as of ggplot2
3.4.0.
i Please use the `linewidth` argument instead.
i The deprecated feature was likely used in the UpSetR package.
Please report the issue to the authors.
```



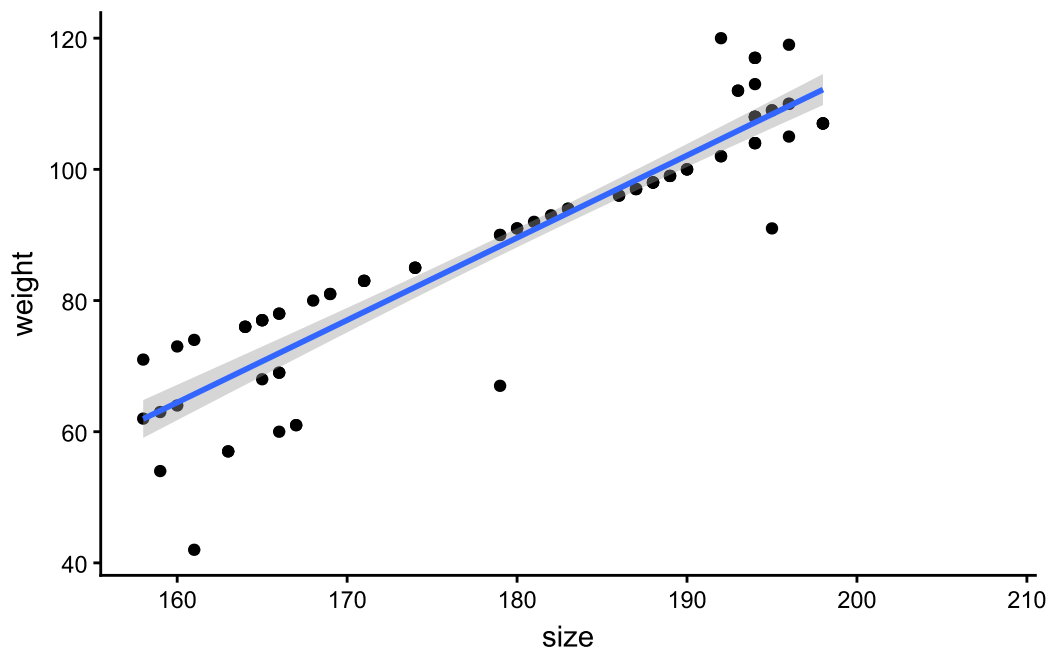
corr lation taille-masse

```
ggplot(snorena) +
 aes(y = weight, x = size) +
 geom_point() +
 geom_smooth(method = "lm") +
 theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

Warning: Removed 14 rows containing non-finite outside the scale range  
(`stat\_smooth()`).

Warning: Removed 14 rows containing missing values or values outside the scale  
range  
(`geom\_point()`).



```
ggsave(
 "img/09Rl27N.png", width = 14, units = "cm"
)
```

Saving 14 x 8.89 cm image  
`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 14 rows containing non-finite outside the scale range  
(`stat\_smooth()`).  
Removed 14 rows containing missing values or values outside the scale range  
(`geom\_point()`).

modèle linéaire

```
lm(weight ~ size, data = snorena) |> summary()
```

Call:  
lm(formula = weight ~ size, data = snorena)

Residuals:

| Min | 1Q | Median | 3Q | Max |
|-----|----|--------|----|-----|
|-----|----|--------|----|-----|

```
-23.7105 -2.5035 0.6022 5.1177 15.3687
```

Coefficients:

```
 Estimate Std. Error t value Pr(>|t|)
(Intercept) -136.42657 9.96946 -13.68 <2e-16 ***
size 1.25551 0.05498 22.83 <2e-16 ***
```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.727 on 84 degrees of freedom

(14 observations effacées parce que manquantes)

Multiple R-squared: 0.8613, Adjusted R-squared: 0.8596

F-statistic: 521.4 on 1 and 84 DF, p-value: < 2.2e-16

remplacement des données manquantes

```
(taille_moyenne <- mean(snorena$size))
```

```
[1] NA
```

```
(taille_moyenne <- mean(snorena$size, na.rm = TRUE))
```

```
[1] 181.0947
```

```
(masse_moyenne <- mean(snorena$weight, na.rm = TRUE))
```

```
[1] 90.3956
```

```
snorena_mod <- snorena |>
 mutate(
 type =
 case_when(
 is.na(weight) ~ "masse moyenne",
 is.na(size) ~ "taille moyenne",
 TRUE ~ "origine"
)
) |>
 replace_na(
 list(
```

```

 size = taille_moyenne |> as.integer(),
 weight = masse_moyenne |> as.integer()
)
)

```

vérification qu'il n'y ait plus de valeurs manquantes

```
snorena_mod |> summary()
```

| age           | weight         | size          | alcohol        | sex     |
|---------------|----------------|---------------|----------------|---------|
| Min. :23.00   | Min. : 42.00   | Min. :158.0   | Min. : 0.000   | M :75   |
| 1st Qu.:43.00 | 1st Qu.: 78.00 | 1st Qu.:166.8 | 1st Qu.: 0.000 | W :22   |
| Median :51.00 | Median : 91.50 | Median :183.0 | Median : 2.000 | NA's: 3 |
| Mean :52.16   | Mean : 90.36   | Mean :181.1   | Mean : 2.905   |         |
| 3rd Qu.:63.00 | 3rd Qu.:104.00 | 3rd Qu.:194.0 | 3rd Qu.: 4.000 |         |
| Max. :74.00   | Max. :120.00   | Max. :208.0   | Max. :15.000   |         |
| NA's :3       |                |               | NA's :5        |         |

| snore   | tobacco | type             |
|---------|---------|------------------|
| N :62   | N :32   | Length:100       |
| Y :32   | Y :61   | Class :character |
| NA's: 6 | NA's: 7 | Mode :character  |

```
snorena_mod |> count(type)
```

|   | type           | n  |
|---|----------------|----|
| 1 | masse moyenne  | 9  |
| 2 | origine        | 86 |
| 3 | taille moyenne | 5  |

graphique

```

snorena_mod |>
 ggplot() +
 aes(y = weight, x = size) +
 geom_smooth(method = "lm") +
 geom_point(aes(color = type, shape = type)) +
 scale_color_manual(values = c("red", "black", "gold")) +

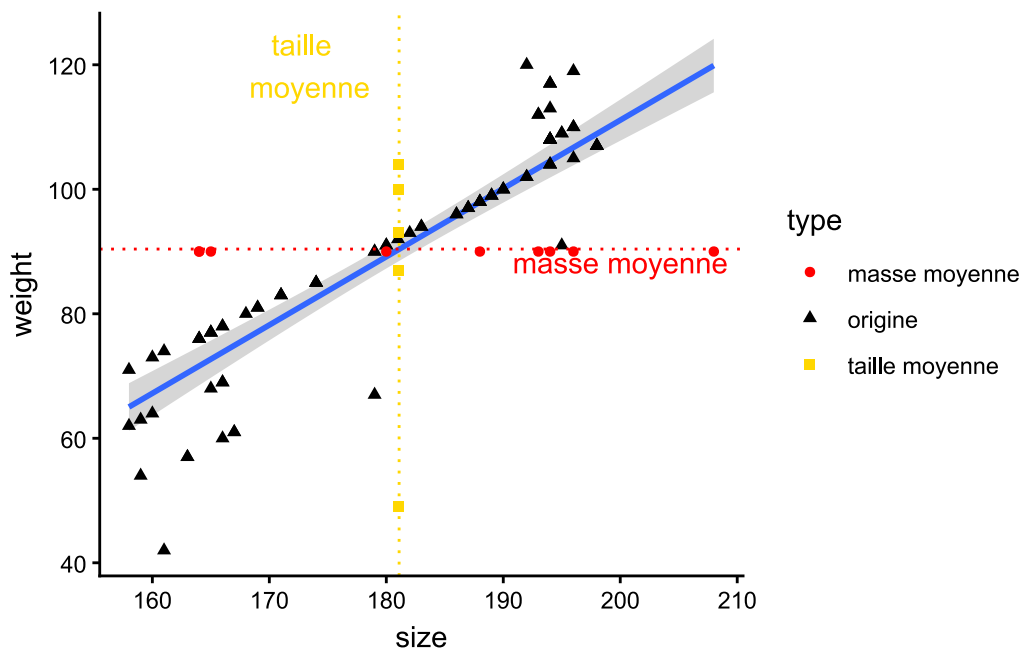
```

```

 annotate("text", x = 200, y = masse_moyenne - 2, label = "masse moyenne", color = "red") +
 geom_hline(aes(yintercept = masse_moyenne), linetype = "dotted", color = "red") +
 annotate("text", x = taille_moyenne - 8, y = 120, label = "taille \n moyenne", color = "gold") +
 geom_vline(aes(xintercept = taille_moyenne), linetype = "dotted", color = "gold") +
 theme_classic()

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```

ggsave(
 "img/09Rl29N.png", width = 14, units = "cm"
)

```

```

Saving 14 x 8.89 cm image
`geom_smooth()` using formula = 'y ~ x'

```

estimation du nombre de dimensions nécessaires

```
(n <- estim_ncpFAMD(snorena))
```



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```
$ncp
[1] 5

$critrion
 0 1 2 3 4 5
66.95078 53.05094 41.28750 34.71121 33.66921 31.86614
```

```
snorena_complete <- imputeFAMD(snorena, ncp = n$ncp)
```

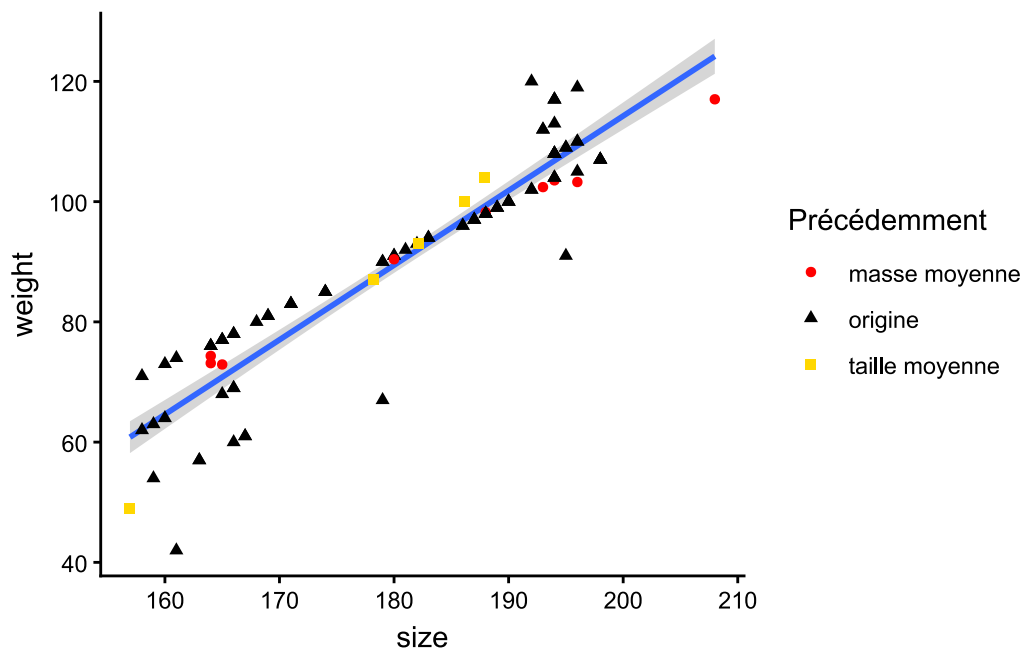
```
snorena_complete$complete0bs |>
 mutate(
```

```

precedement = snorena_mod$type
) |>
ggplot() +
aes(y = weight, x = size) +
geom_smooth(method = "lm") +
geom_point(aes(color = precedement, shape = precedement)) +
scale_color_manual(values = c("red", "black", "gold")) +
labs(color = "Précédemment", shape = "Précédemment") +
theme_classic()

```

`geom\_smooth()` using formula = 'y ~ x'



```

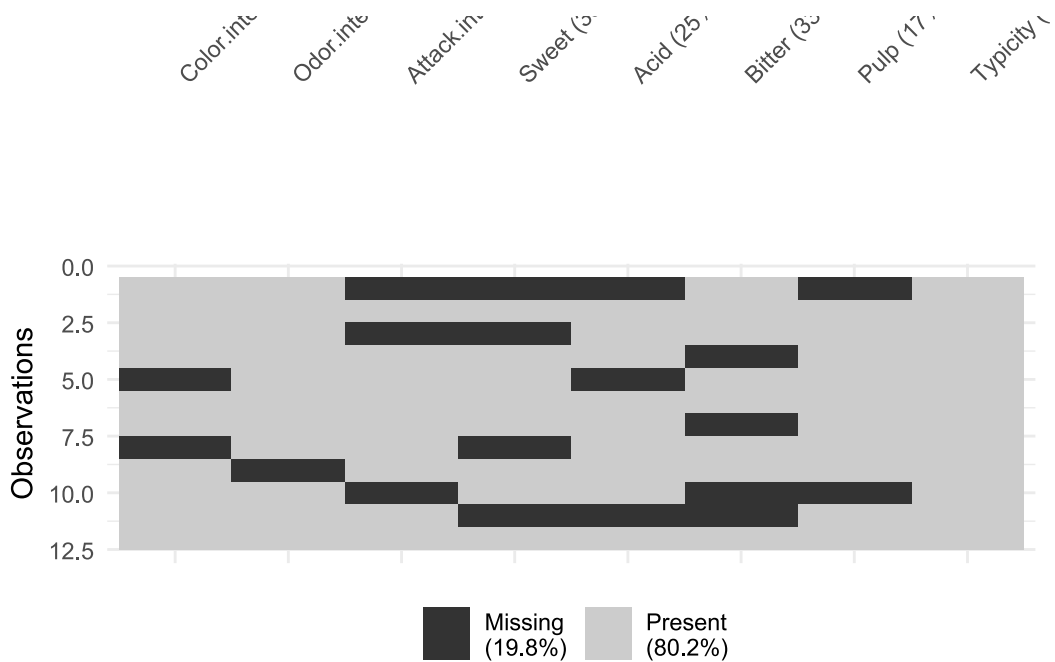
ggsave(
 "img/09Rl30N.png", width = 14, units = "cm"
)

```

Saving 14 x 8.89 cm image  
`geom\_smooth()` using formula = 'y ~ x'

test jeu de données orange

```
data("orange")
visdat::vis_miss(orange)
```



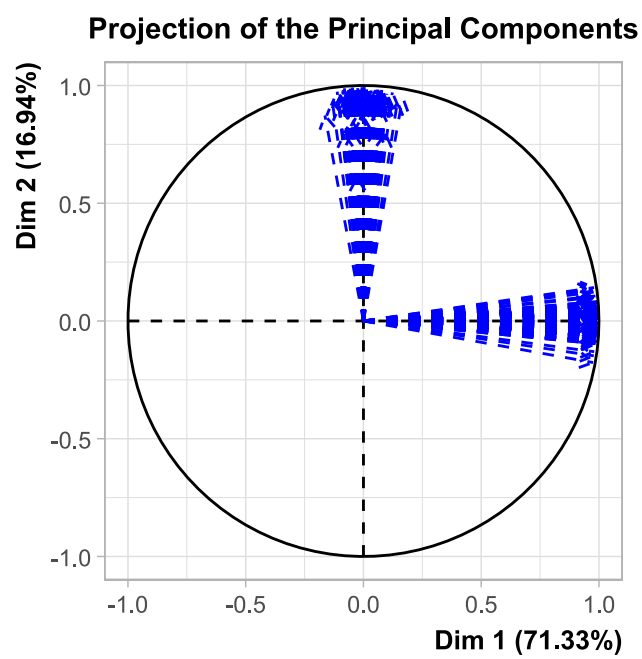
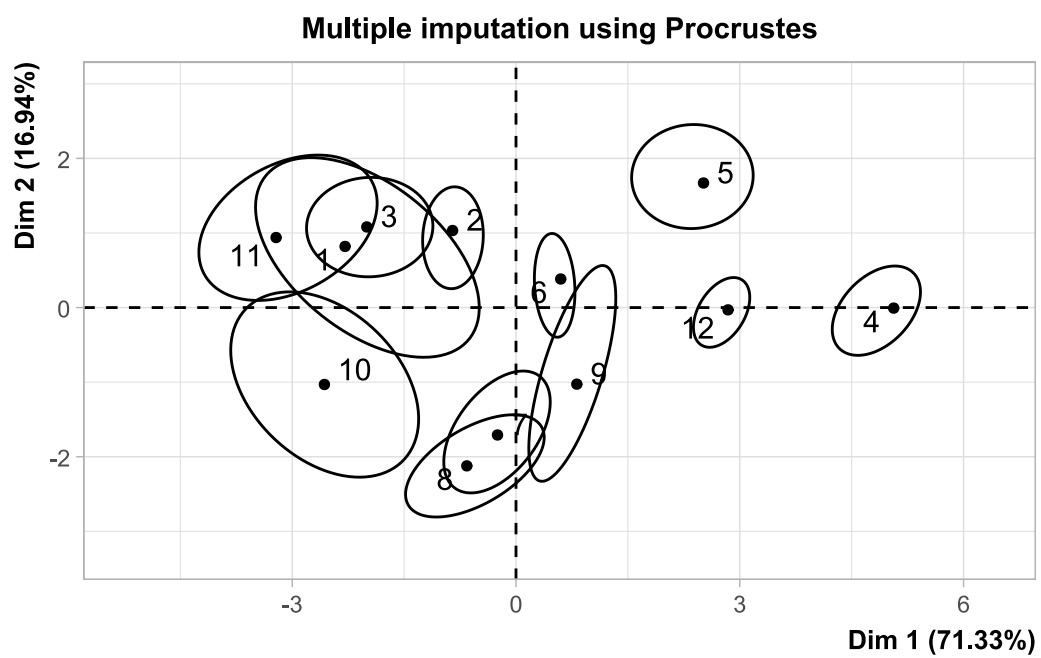
imputation multiple

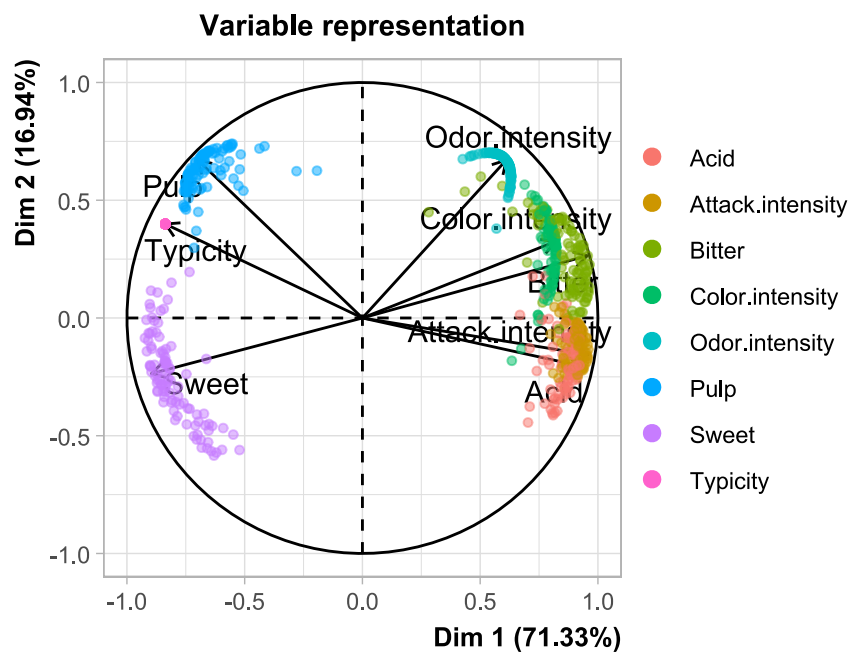
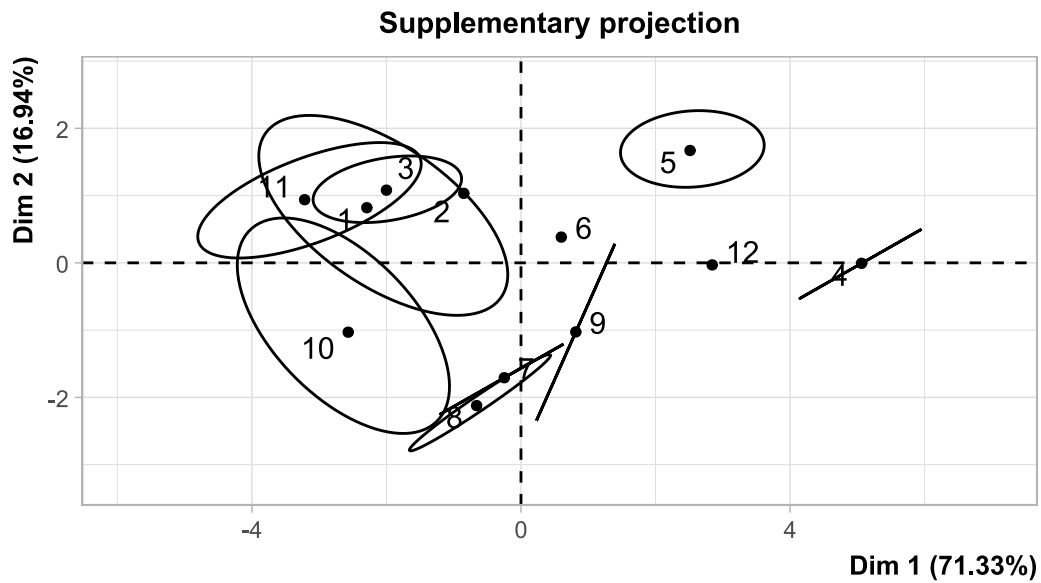
```
orange_imputation_multiple <- MIPCA(orange, ncp = estim_ncpPCA(orange)$ncp)
```

```
Warning in impute(X, ncp = ncp, scale = scale, method = method, ind.sup =
ind.sup, : Stopped after 1000 iterations
```

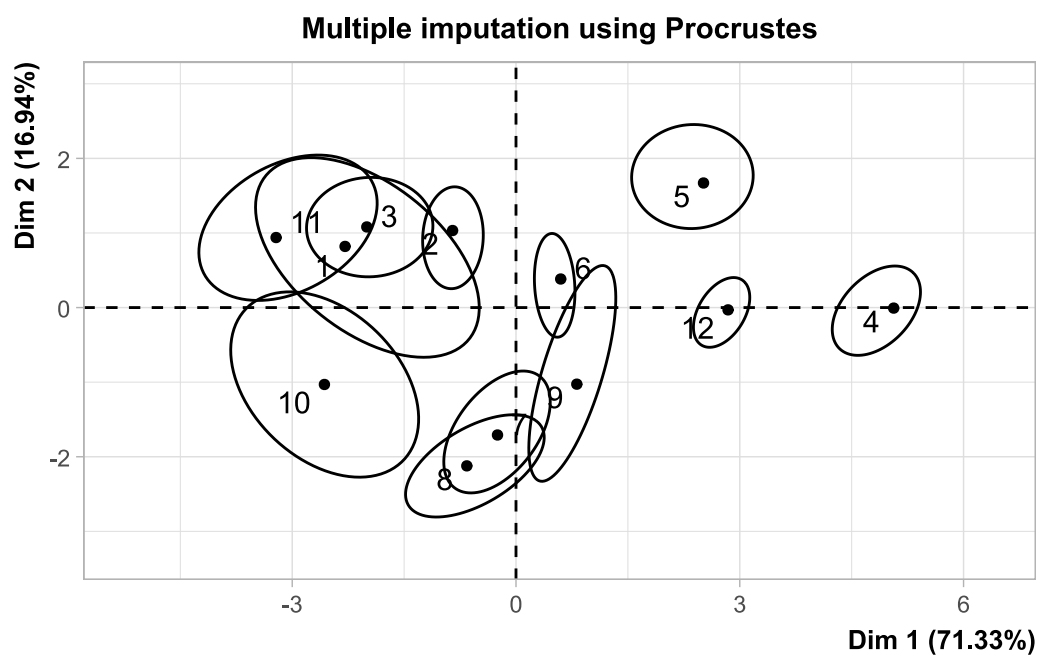
représentation graphique

```
plot.MIPCA(orange_imputation_multiple)
```

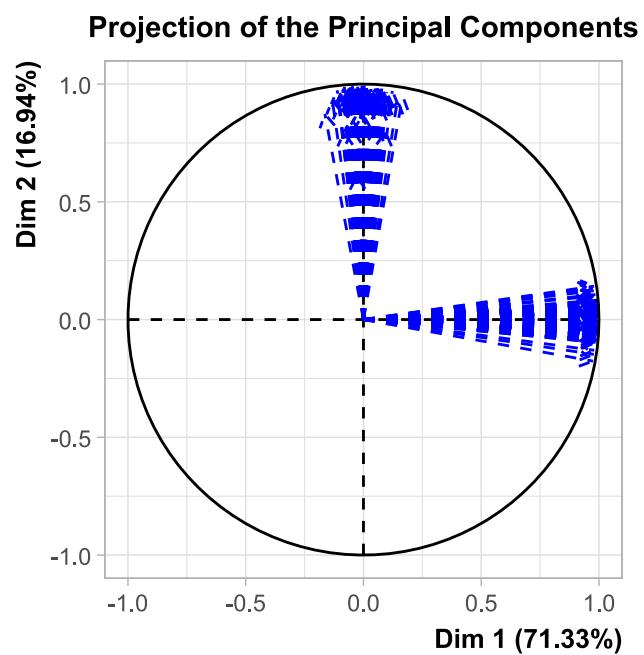




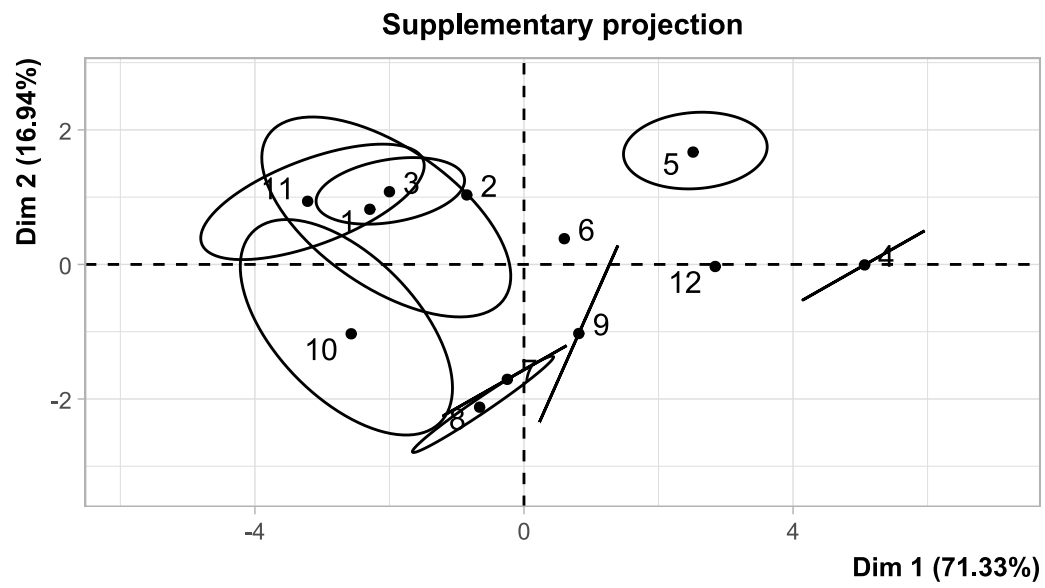
`$PlotIndProc`



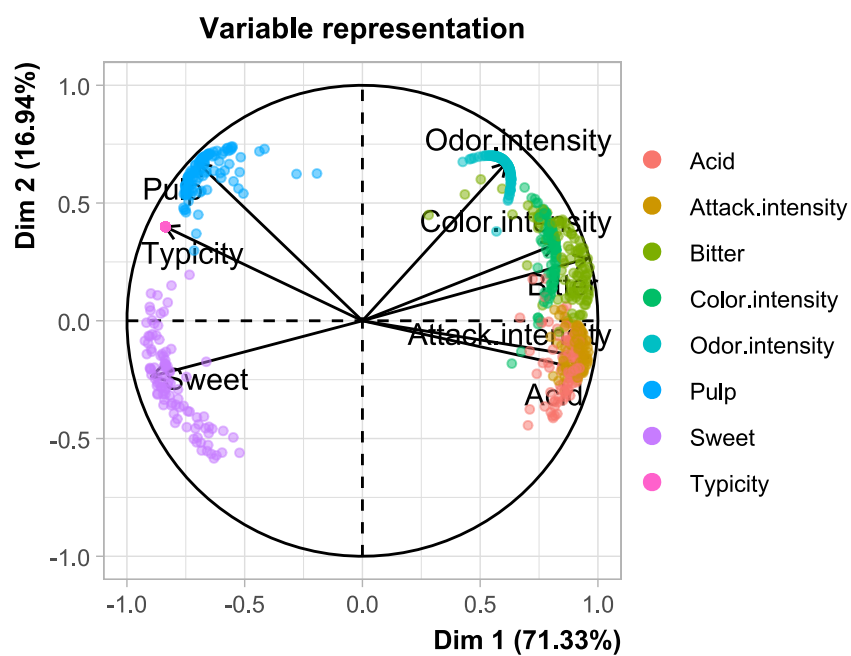
`$PlotDim`



\$PlotIndSupp



\$PlotVar



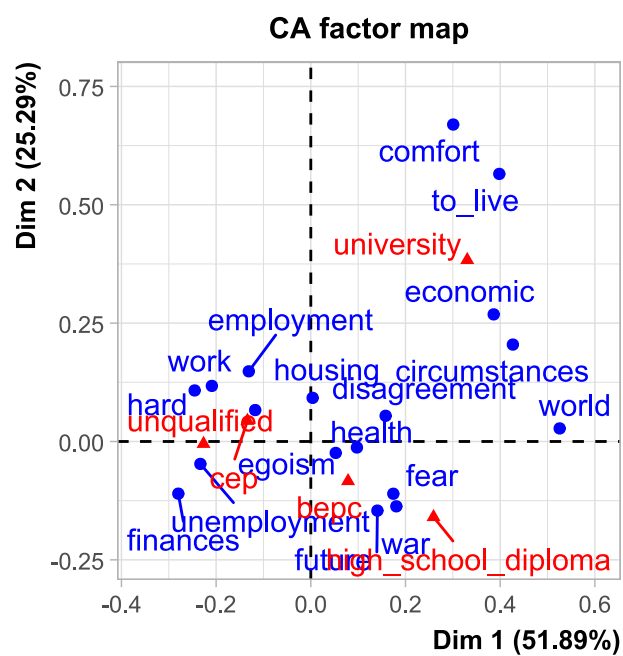
## 5 AFC

```
data("children")
envie_enfant <- children |>
 select(- "thirty", - "fifty", - "more_fifty")
```

représentation graphique

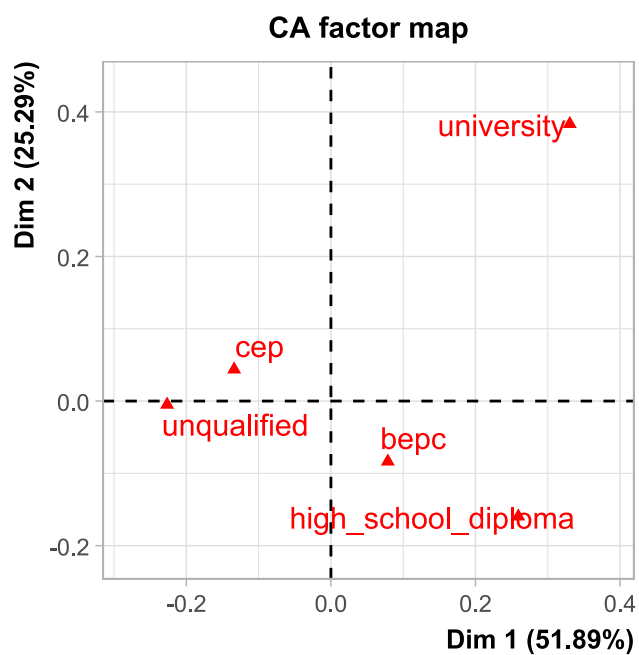
```
afc <- CA(envie_enfant)
```

Warning: ggrepel: 1 unlabeled data points (too many overlaps). Consider increasing max.overlaps

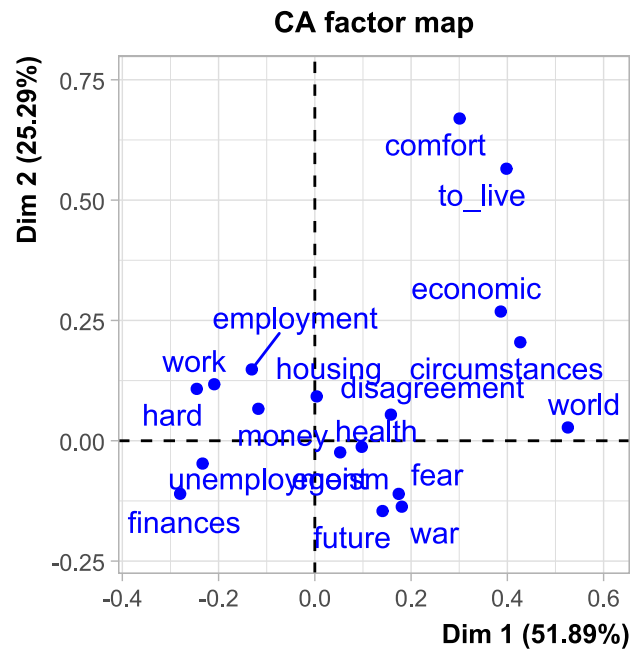


seulement les lignes ou les colonnes

```
plot.CA(afc, invisible = "row")
```



```
plot.CA(afc, invisible = "col")
```



```
ellipseCA(afc, ellipse = "col")
```

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_point()`).

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_text\_repel()`).

Warning: ggrepel: 1 unlabeled data points (too many overlaps). Consider increasing max.overlaps

